

Master's thesis

Master's Programme in Particle Physics and Astrophysical Sciences

Group finding algorithms for galaxy cluster substructure detection from spectroscopy

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Tiivistelmä — Referat — Abstract

Galaxy clusters and their mergers are irreplaceable probes for the large-scale structure of the universe and structure formation, while the intra-cluster environment contains information about galaxy interactions and group dynamics. In order to resolve these environments in three dimensions, group finding algorithms are used with spectroscopic redshift information of cluster members. This work provides a new analysis of optical member catalogs, algorithm performance, and recovery of substructure properties from simulations.

For optical data, the spectroscopic redshift ($z_{\rm spec}$) completeness of the DECaLS redMaPPer catalog is evaluated. The utility of catalog enrichment from novel spectroscopic data from DESI DR1 is tested for the purpose of substructure analysis. Four group finding algorithms are then investigated for their efficacy for subcomponent detection. Further analysis is conducted with the Gaussian Mixture Modeling algorithm `mclust`. It is applied to mock data from the Uchuu simulation, with a focus on rich ($\lambda > 30$) subcomponents due to their relevance for the determination of dynamical properties. The influence of $z_{\rm spec}$ completeness is tested by reducing completeness artifially between the range $10\% < c < 100\%$. Clustering solutions are assessed by examining five Cluster Validity Indexes, and comparing to results from photometric data. Finally, the recovery of the velocity dispersions (σ_v) and virial masses ($M_{\rm vir}$) of subcomponents is examined.

These investigations find that optical catalog enrichment with a larger and fainter galaxy catalog is a credible stategy, resulting in over 75% improvement in the number of clusters in the high-completeness regime. Comparison of substructure detection with `mclust` with a photometric study also suggests that clustering solutions from spectroscopy outperform photometry above $c \sim 20\%$. However, no significant improvements were seen with increasing completeness in the range $50\% < c < 100\%$. The detected substructures exhibit a consistent underestimation of σ_v, reaching fractions of ~ 0.8 of the true subhalo members. Results are placed in comparison with the $\sigma_v - M_{\rm vir}$ relation from previous literature, which reveals that massive subcomponents are more significantly affected by the low σ_v measurements.

These results suggest a requirement for re-calibration of the $\sigma_v - M_{\rm vir}$ relation for substructure, and raise the importance of statistical studies of group finders, the lack of which represents a current gap in the field.

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galaxies: clusters: general, galaxies: kinematics and dynamics, haloes, methods: data analysis

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"The universe is a big place, perhaps the biggest."
— From *Venus on the Half Shell* by Philip José Farmer

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1. Introduction

Galaxy clusters are the largest gravitationally bound objects in the universe, making their observations irreplaceable in fields of research concerning its evolution and large-scale structure. Because of their abundance in cosmological space and time scales, extensive surveys of galaxies and their clustering allow us to place constraints on cosmological model parameters, properties of dark energy and large-scale structure evolution. Within clusters, measuring the distribution of member galaxies reveals information about how galaxies populate dark matter haloes, and more generally about the galaxy-halo connection (see Wechsler and Tinker, 2018 for a review).

There is also an incentive to formally study the galaxy environment for improving our understanding of galaxy formation and evolution. Dark matter plays a large role in initializing galaxy formation, but an abundance of other phenomena additionally affect galaxies during and after their formation (Voit, 2005; Tempel et al., 2017).

Studying properties of clusters at different epochs allows us to study their formation history, as well as discover regularities and correlations. Particularly, as the current paradigm describes structure formation as hierarchical, understanding and identifying merger events becomes crucial for studying cluster evolution. Nonetheless, the number of thoroughly studied substructure detection methods remains small, and physical processes underlying mergers are still poorly understood. Tempel et al. (2017) emphasize this point by stating that "there is thus a need and appetite in the community for a combination of the targeted studies that can successfully identify individual merging events and the large-scale grouping algorithms that can estimate the statistics of clustering among galaxies." In particular, detecting mergers from galaxy redshift measurements presents valuable information about the dynamical states of clusters, but many methods continue to suffer from limitations that are poorly adressed (de Los Rios et al., 2016).

This thesis aims to contribute to filling this gap by evaluating the performance of four group finding algorithms for the purpose of detecting galaxy cluster mergers by testing them with a simulated optical cluster catalog and investigating the dynamical properties of substructures. In the current times where both the scale and quality of large spectroscopic surveys are drastically improving (e.g. DESI, LSST and PFS),

testing the performance of statistical analysis directly impacts our ability to efficiently utilize the extent of new surveys. Necessarily, this also highlights the question about possible biases and limitations of redshift survey data itself, which is discussed in this work by analysis and application of previous literature and novel spectroscopic data.

The structure of this work is as follows. Sections 1.1 and 1.2 introduce the relevant cosmological background as well as the observable properties and mass proxies that will be utilized throughout this study. Section 2 describes the simulated and observational datasets, and the functionalities of the group finding algorithms. Statistical and dynamical results of the group finder runs are presented and compared to literature in Section 3. The work is concluded by discussing and summarizing algorithm performances as well as data utilization, their limitations, and future prospects in Section 4.

1.1 Cosmology

The current most popular cosmological model is the ΛCDM model, which describes a flat universe with the following three componets: dark energy (Λ), collisionless "cold" dark matter (CDM), as well as a baryonic matter component. While different versions of the ΛCDM model are used, the measurable (independent) cosmological parameters at the core of the Planck collaboration report (Planck Collaboration et al., 2020) are dark matter density $\Omega_c h^2$, baryon density $\Omega_b h^2$, scalar spectral index n_s, optical depth τ, and angular acoustic scale $100\theta_*$, while inferred (model-dependent) parameters include the Hubble constant H_0, matter density parameter Ω_m, and matter fluctuation amplitude σ_8.

Galaxy clusters are used for cosmological studies in a variety of methods. As they are the largest gravitationally bound objects whose mass we can estimate due to virial equilibrium, it is possible to compare mass distributions at different time scales and shed light on the rate of structure formation (Voit, 2005).

The distribution of galaxies has been used to probe the cosmic density field by using them as tracers of the distribution of mass in the Universe. However, this is only a rough estimation, as galaxies only make up a small fraction of the total mass distribution—discounting e.g. the intracluster medium (ICM) and dark matter (DM)—and because the spatial distribution of galaxies is also dependent on their individual properties (Mo et al., 2010, p. 262). Even so, clusters are irreplaceable probes for cosmological studies, as their different constituents can be detected across wide wavelength ranges (Myles et al., 2025).

One of the most valuable, yet challenging, properties to estimate is the cluster mass, as it cannot be observed directly. As a result, other probes are needed, such as

the peculiar velocity distribution of galaxies, which will be discussed in more detail in the next subsections.

Throughout this work, a cosmology with $\Omega_m = 0.3$, $\Omega_\Lambda = 0.7$, and $H_0 = 100\ h$ $\mathrm{kms}^{-1}\mathrm{Mpc}^{-1}$, where $h = 0.7$ is assumed, unless otherwise specified.

1.2 Mass proxies

1.2.1 Optical richness

Cluster optical richness, λ, is a measure of the galaxy content of a given cluster. It is not necessarily the absolute number of galaxies located within the cluster, but an estimator or tracer of optical mass. There is a variety of estimation methods for λ, but one of the most common is the probabilistic approach laid out by Rozo et al. (2009) and Rykoff et al. (2012), which is applied in this work as well.

Defining this estimator begins with modeling the projected galaxy distribution using $S(\mathbf{x}) = \lambda u(\mathbf{x}|\lambda) + b(\mathbf{x})$, where $\mathbf{x}$ is the observable property vector of a galaxy, λ is the number of galaxies, $u(\mathbf{x}|\lambda)$ is the normalized galaxy density profile, and $b(\mathbf{x})$ is the background galaxy density.

Then, the probability that a given galaxy in the cluster's region in the plane-of-sky is a member of the cluster is given from the ratio of the estimated galaxy distribution of the cluster and the combined distribution of the cluster and background:

$$p(\mathbf{x}) = \frac{\lambda u(\mathbf{x}|\lambda)}{\lambda u(\mathbf{x}|\lambda) + b(\mathbf{x})}. \tag{1.1}$$

In addition, the richness estimate is required to fulfill a constraint equation summing over all galaxies in the cluster field within a chosen member count-dependent cutoff radius $R_c(\lambda)$:

$$\lambda = \sum p(\mathbf{x}) = \sum_{R<R_c} \frac{\lambda u(\mathbf{x}|\lambda)}{\lambda u(\mathbf{x}|\lambda) + b(\mathbf{x})}. \tag{1.2}$$

Notably, the only unknown in the above equation when working with observational data is λ. It can then finally be solved for e.g. by using a zero-finding algorithm (Rykoff et al., 2012, 2014).

1.2.2 Velocity dispersion

Peculiar velocities of galaxies in clusters obtained from spectroscopic redshift measurements are widely used in the literature, and are considered precise tracers for the underlying mass distribution of haloes and clusters, referred to as the dynamical mass.

In the low-redshift regime covered in this work, $z \lesssim 0.2$, the redshift – velocity relation can be approximated by the linear expression $z = v/c$. It is, however, useful to remember that the true correlation between recessional velocity and redshift is not linear, but dependent on the scale factor of expansion, and hence depends on the specific cosmological model used.

A commonly used method for calculating the velocity dispersion from galaxy velocity information is the Gapper method (Beers et al., 1990), described by the following equation (following notations in Eckert et al., 2017):

$$\sigma_{\text{Gapper}} = \frac{\sqrt{\pi}}{N(N-1)} \sum_{i=1}^{N-1} \Delta v_i w_i, \tag{1.3}$$

where N is the number of galaxies in the group (in this case the number of galaxies with velocity data available), Δv_i are the velocity gaps, i.e., $v_{i+1} - v_i$, and w_i are the weights, given by $i(N-1)$.

The Gapper estimate uncertainties are computed as described in Ruel et al. (2014), given by the expression

$$\Delta\sigma_{\text{Gapper}} = \frac{C_{\text{Gapper}}\sigma_{\text{Gapper}}}{\sqrt{N-1}}, \tag{1.4}$$

where C_{Gapper} is a constant for which the value reported by Ruel et al., $C_{\text{Gapper}} = 0.91$, is used throughout this work.

1.2.3 Virial mass

The virial mass of a cluster is another mass estimate, defined based on the assumption that the virial theorem holds within a specified region of the cluster. First, the virial radius is defined as the radius within which the density is the critical density of the universe, ρ_c, multiplied by an overdensity coefficient, Δ_c (Mo et al., 2010). The critical density is obtained using the time-dependent Hubble parameter, $H(t)$, and the gravitational constant, G, as $\rho_c = 3H^2(t)/8\pi G$. The virial mass is then simply calculated from the mass contained within the virial radius:

$$M_{\text{vir}} = \frac{4}{3}\pi r_{\text{vir}}^3 \rho(< r_{\text{vir}}) = \frac{4}{3}\pi r_{\text{vir}}^3 \Delta_c \rho_c, \tag{1.5}$$

where r_{vir} is the virial radius and $\rho(< r_{\text{vir}})$ is the density within r_{vir}. The most commonly used value for the overdensity factor is $\Delta_c = 200$, and will be adopted in this work unless otherwise specified.

Obtaining estimates for the virial mass or radius from observations is generally done from comparisons to models, commonly the NFW profile for the structure of dark matter haloes.

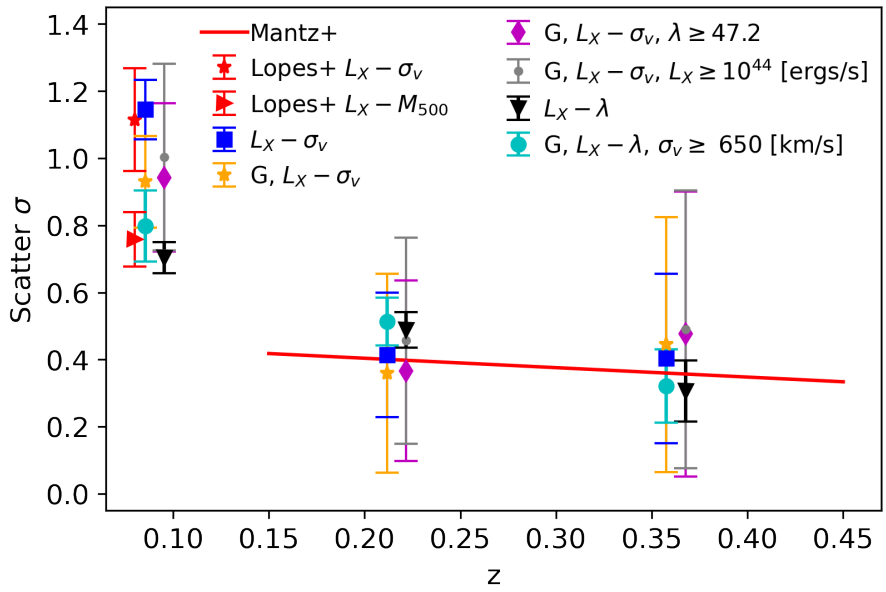


Figure 1.1: X-ray luminosity L_X vs. optical mass proxy scaling relations. From Damsted, S., Finoguenov, A., Clerc, N., Davalgaitė, I., Kirkpatrick, C. C., Mamon, G. A., Ider Chitham, J., Kiiveri, K., Comparat, J., & Collins, C. (2023). CODEX: Role of velocity substructure in the scaling relations of galaxy clusters. *Astronomy and Astrophysics*, *676*, Article A127. https://doi.org/10.1051/0004-6361/202245308 p. 14.

1.2.4 Scaling relations

Researching the physical processes involved in cluster formation and evolution are generally studied using scaling relations between observational and dynamical properties, including σ_v, $M_{\rm vir}$, λ, and in many cases X-ray luminosity, L_X. Studying the correlation between these properties and how well they are constrained, i.e., quantifying the scatter found around the relations, points out questions about processes not yet understood.

One such question is around the fact that increased scatter is consistently found for a multitude of scaling relations in the low-redshift regime $z \sim 0.1$. Figure 1.1, applied from Damsted et al. (2023), exemplifies this effect and shows the scatters recorded by different studies for L_X–σ_v, L_X–M_{500}, and L_X–λ relations evolving with redshift.

Answering the questions about why this scatter is seen is a current topic in the field, and the role of substructure in this observed scatter has been discussed in numerous studies, with contradictory results (see Lopes et al., 2009, and references therein). Because substructure is a sign of incomplete relaxation (non-cool core) and seen both in ICM X-ray emission and as non-Gaussian distribution of member galaxies, it would be expected to find that improved sub-cluster identification and individual consideration to the scaling relations could affect the scatter.

2. Data and methods

2.1 Optical data

2.1.1 redMaPPer Galaxy Cluster Catalog

The main optical data catalog studied in this work is a sub-catalog from the Dark Energy Camera Legacy Survey (DECaLS), which is part of the DESI Legacy Imaging Survey (Dey et al., 2019), from which optical clusters are identified using the redMaPPer (red-sequence Matched-filter Probabilistic Percolation) algorithm (Rykoff et al., 2014).

RedMaPPer is a photometric cluster finding algorithm focused on old, red galaxies and photometric redshift estimates, assigning members probabilities and estimating cluster richness from the sum of these probabilities as discussed in Section 1.2.1 (Rykoff et al., 2014). The optical DECaLS data is processed using eROMaPPer, an enhanced redMaPPer package developed by Kluge et al. (2024), for use alongside eROSITA data.

The catalog is implemented in this work as part of examining data utilization strategies for substructure analysis, by creating and employing a suitable sub-catalog. While redMaPPer is compiled on photometric data, the catalog supplies spectroscopic redshift measurements for galaxies from previous literature, when available.

This spectroscopic data availability raises a key subject in substructure analysis from galaxy members, namely the question between photometric data and redshift (z_{photo}) and spectroscopic redshift (z_{spec}). Photometric redshift measurements are generally too imprecise for significant intracluster analysis, so substructure identification from photometric data is often limited to 2-dimensional analysis, losing information in the velocity space. Spectroscopy allows for 3-dimensional mapping of the galaxy distribution in a cluster, but is often severely completeness-limited, consequently making z_{spec} completeness in the catalog a relevant factor of consideration within this study. Completeness refers to the fraction of member galaxies within a cluster with available z_{spec} measurements.

Multiple cuts on the cluster catalog variables are imposed to filter for clusters suitable for substructure analysis. This work focuses on a redshift range of $z \leq 0.2$,

and a lower richness limit is set to $\lambda \geq 14$ because meaningful substructure detection and study of dynamical properties is not expected for poorer clusters. Additionally, cuts in the "scale factor" and "mask fraction" are made.

The scale factor S (*scaleval*) and mask fraction $f_{\rm mask}$ (*maskfrac*) are computed in the catalog in order to correct for cluster members missing due to limiting magnitude and masking, respectively (Rykoff et al., 2014). In other words, S describes how many galaxies have magnitudes above the survey limit. For the case of DECaLS, the median 5σ detection limit for galaxies in g-band AB magnitude is $m_{g,lim} = 23.72$. The following expression is obtained for the richness (compare difference to Equation 1.2):

$$\frac{\lambda}{S} = \sum_{\rm gals} p_{\rm mem}, \tag{2.1}$$

where $p_{\rm mem}$ is the membership probability of the given galaxy. Therefore S is a function of redshift, and can be expressed as follows:

$$S(z) = \frac{1}{1 - \langle C|z \rangle}, \tag{2.2}$$

where $\langle C|z \rangle$ is the expectation value of C, which is another correction factor dependent on galaxy luminosity, position, and the number of galaxies in given pixels.

The mask fraction $f_{\rm mask}$ describes all other masking except the magnitude limit, namely geometric factors such as stars, bad fields or edge effects. It is hence a measure of the effective area of the cluster that has been masked. It is defined as

$$f_{\rm mask} = \frac{\int_{\rm out} u(x)dx}{\int u(x)dx}, \tag{2.3}$$

where x describes the position in the observable space, $u(x)$ is the probability distribution for x, and the integral $\int_{out}$ integrates over the positions outside the mask. The mask fraction values reported in the redMaPPer catalog are evaluated using Monte Carlo integration.

The preliminary cuts made to the catalog data to prepare it for subcluster analysis are then $z \leq 0.2$, $\lambda \geq 14$, $S < 1.4$, and $f_{\rm mask} < 0.2$, yielding a cluster catalog with 5823 clusters.

However, $z_{\rm spec}$ completeness remains a challenge for analysis, as the number of members available as input for 3-dimensional group finding algorithms is greatly reduced. To alleviate the issue, the chosen redMaPPer clusters are matched to, and enriched with, additional $z_{\rm spec}$ values from a new large-scale spectroscopic survey. This is described in the next subsection.

2.1.2 DESI Data Release 1 (DR1)

Great improvements have been seen in the availability of spectroscopic data in the field in recent years, particularly noting the significance of the Dark Energy Spectroscopic Instrument (DESI) collaboration (DESI Collaboration et al., 2026) spectroscopic redshift survey, which includes over 18 million redshift measurements. The instrument is mounted on the 4m Mayall Telescope at Kitt Peak National Observatory (KPNO) in Arizona, USA.

For this study, galaxy member $z_{\rm spec}$ data from DESI is projected onto the selected set of the redMaPPer catalog. This is done in order to obtain better-behaved and pre-grouped data compared to running group finders on DESI data alone. Instead, enriching the existing redMaPPer cluster member catalog with new DESI $z_{\rm spec}$ data allows for utilizing improved completeness while avoiding challenges regarding DESI's sampling function.

Projection is done by coordinate-based matching on the celestial sphere, where only matches with separations under `sep` < 2 [arcsec] were kept. This is then followed by redshift enrichment, where DESI spectroscopic redshifts are assigned to matched members with no $z_{\rm spec}$ available from previous literature. In cases where a redshift is reported both in redMaPPer and the DESI catalogue, DESI redshift is preferred.

These values are also compared to study the agreement between DESI and previous literature, namely by computing the redshift scatter. A total of 1761 members were found with $z_{\rm spec}$ values reported both in redMaPPer and DESI DR1, from which residuals are calculated as $r = |(z_{\rm DESI} - z_{\rm lit})/(1 + z_{\rm lit})|$, where $z_{\rm lit}$ is the spectroscopic redshift value reported in the redMaPPer member catalog and $z_{\rm DESI}$ is the new value supplied by DESI DR1. Figure 2.1 shows the distribution of the logarithmic residuals given by $r_{log} = \log_{10}|(z_{\rm DESI} - z_{\rm lit})/(1 + z_{\rm lit})|$.

Redshift comparison is additionally checked for outliers, which become apparent near $r_{log} \sim 0$ in Figure 2.1. An outlier fraction of 1.01% is found, and removed from further $z_{\rm spec}$ matching analysis. With outliers removed, the mean of the residuals is $\overline{r} = 2.04 \times 10^{-5}$. The order of magnitude of this value, which corresponds to ~ 6 km s^{-1} in velocity space, is overall significantly under the threshold for being critical for cluster velocity dispersions (generally of the order $\sim 10^2$ km s^{-1}).

Additionally, object magnitude inevitably influences the spectroscopic redshift measurements due to increased noise with magnitude. This is illustrated in Figure 2.2, which depicts the probability density of the residuals for different magnitude bins. As expected, the residual distribution skews larger as magnitude increases. Particularly, as magnitudes exceed mag $= 17$ a new peak in the distribution forms around $r_{log} \approx -3$, which is now notably in a similar order of magnitude as cluster velocity dispersions.

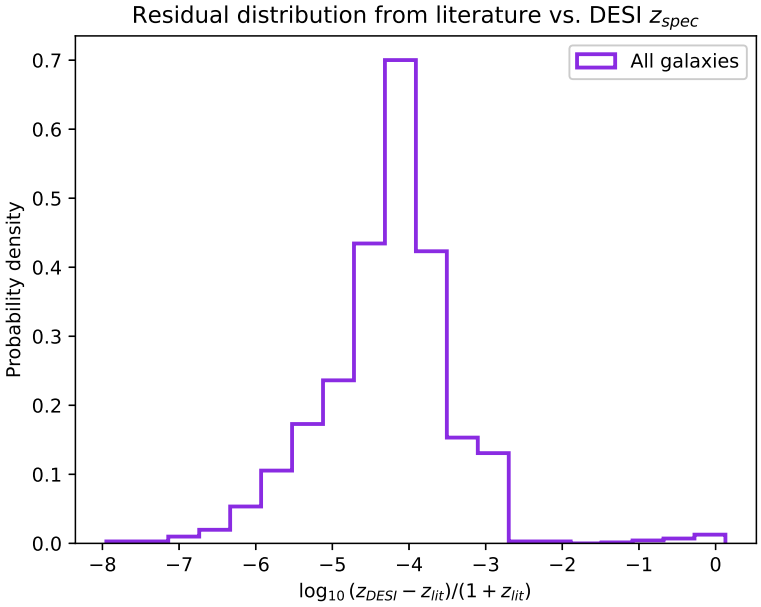


Figure 2.1: Number distribution of the logarithmic residuals between literature and DESI $z_{\rm spec}$ in the defined subset, given by $r_{log} = \log_{10}|(z_{\rm DESI} - z_{\rm lit})/(1 + z_{\rm lit})|$. The plot shows a clear peak at $r_{log} \approx -4$. Outliers are observed near $r_{log} \approx 0$, and are removed from subsequent analysis.

Hence, this area of the distribution warrants further study. Figure 2.3 shows the fractions of matches for which the residuals satisfy $\log_{10}|(z_{\rm DESI} - z_{\rm lit})/(1 + z_{\rm lit})| > c$, where both $c = -3$ and $c = -4$ are presented. While across all magnitude bins the fractions of values with $r_{log} > -3$ remains significantly smaller than $r_{log} > -4$, the increase of residual magnitude is apparent with increasing magnitude, as expected.

The final optical cluster catalog consists of 5823 redMaPPer clusters, and a total of 51158 associated galaxies in the member catalog. There are a total of 9589 spectroscopic redshifts available for these members combined from redMaPPer and DESI, whereas the redMaPPer only $z_{\rm spec}$ availability was at 7188 members, improving the spectroscopic redshift completeness of the whole dataset from 14.1% to 18.7%. Figure 2.4 represents the spectroscopic redshift completeness distribution for the sample of redMaPPer clusters before and after spectroscopic redshift enrichment from DESI. The completeness is calculated as $N_{z_{\rm spec}}/N_{\rm gal}$, where $N_{z_{\rm spec}}$ is the number of galaxies in a cluster with $z_{\rm spec}$, and $N_{\rm gal}$ is the total number of member galaxies. Here it is apparent that the overall number of high-completeness clusters in the set has been increased significantly. However, assessing whether the amount of data suitable for statistically meaningful analysis has been improved additionally depends on the absolute number of available redshifts per cluster. Namely, a completeness fraction of 0.1 could refer to one redshift for a cluster of richness 10, or 10 clusters for a large $\lambda = 100$ cluster. Despite identical completeness, the cluster with only one member cannot be utilized

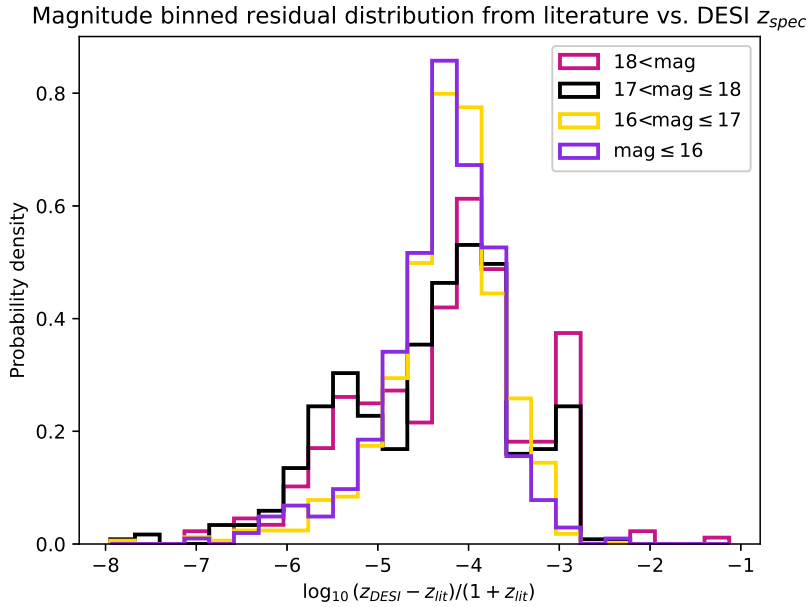


Figure 2.2: Probability density distribution of the logarithmic residuals between literature and DESI $z_{\rm spec}$, given by $\log_{10}|(z_{\rm DESI} - z_{\rm lit})/(1 + z_{\rm lit})|$ within different magnitude bins. Objects with magnitudes mag $>$ 18 are shown in burgundy, magnitudes $17 <$ mag ≤ 18 in black, magnitudes $16 <$ mag ≤ 17 in yellow, and magnitudes mag ≤ 16 in violet.

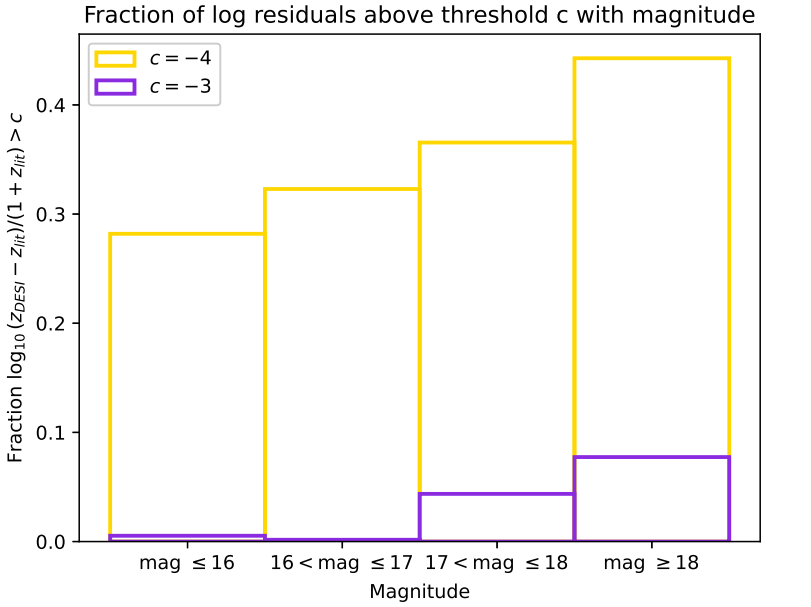


Figure 2.3: The fraction of residuals for which $\log_{10}|(z_{\rm DESI} - z_{\rm lit})/(1 + z_{\rm lit})| > c$, shown for the same magnitude bins as in Figure 2.2. Yellow bins show the fraction at $c = -4$, and violet corresponds to $c = -3$.

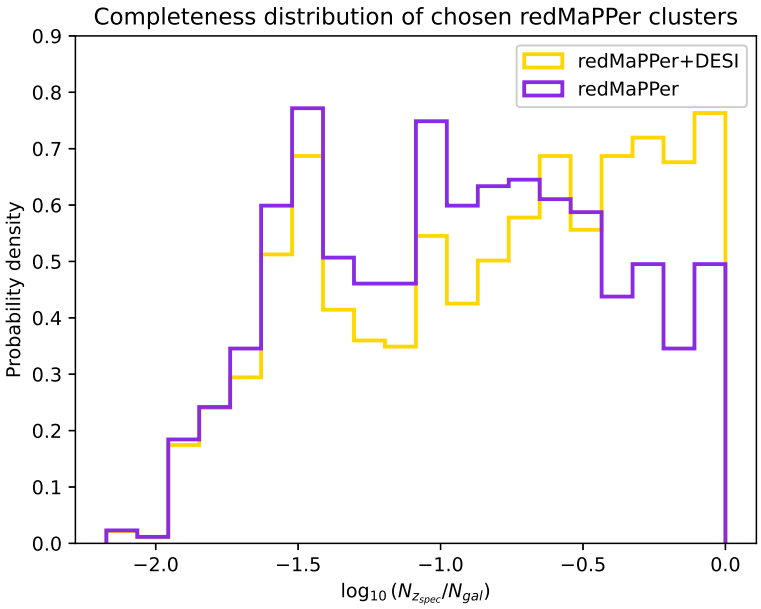


Figure 2.4: Completeness distribution for the previous literature spectroscopic redshifts from redMaPPer (violet) and after DESI enriching (yellow) of the selected clusters ($z \leq 0.2$, $\lambda \geq 14$, scaleval ≤ 1.4, $f_{\rm mask} < 0.2$). The completeness is calculated as the ratio between the number of $z_{\rm spec}$ measurements available for a cluster versus the total number of member galaxies, $N_{z_{\rm spec}}/N_{\rm gal}$. Clusters with 0 spectroscopic members are omitted from the plot.

for substructure analysis at all.

This issue is investigated by counting up the number of available $z_{\rm spec}$ values per cluster and studying the distribution, the results of which are shown in Figure 2.5. The overall distribution has skewed towards higher counts, where improvements are specifically seen in the range $N_{spec} > 25$, hence also increasing the number of clusters viable for significant results in subclusture detection.

Thus, advancements in completeness are found accross cluster sizes by utilizing DESI enriched data. Despite improved completeness, advances with cluster coverage using DESI remain limited with Data Release 1. For this reason, theoretical methods are needed for characterizing cluster substructure detections.

2.2 UCHUU Simulation Data

Simulation data is utilized in order to assess the performance of different 3-dimensional group finding algorithms for improving cluster substructure analysis. Simulations have the advantage that groups reconstructed by algorithms can be compared to the known haloes given in the catalogs, as opposed to working directly on observational data.

For this purpose, mock data from Data Release 1 of the Uchuu simulation suite of

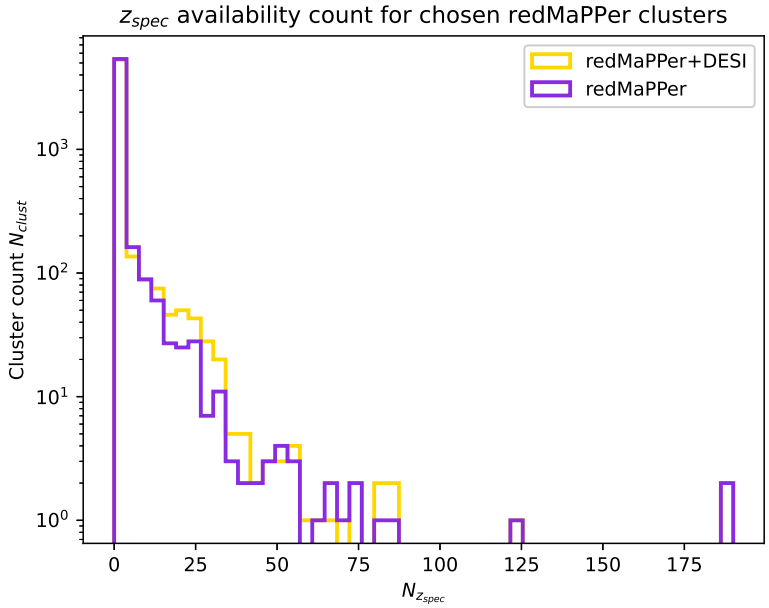


Figure 2.5: Distribution for spectroscopic redshift counts per cluster ($N_{z_{\rm spec}}$) for chosen redMaPPer clusters ($z \leq 0.2$, $\lambda \geq 14$, scaleval ≤ 1.4, $f_{\rm mask} < 0.2$) before and after DESI enrichment in violet and yellow, respectively. Clusters with 0 spectroscopic members are included in the plot by setting completeness as an arbitrary small positive number.

ultra-large DM-only cosmological N-body simulations is used (Ishiyama et al., 2021). The simulations included in the Uchuu DR1 suite together consist of over 2 trillion dark matter particles, box sizes varying from 2.0 h^{-1}Gpc to 100.0 h^{-1}Mpc and high particle mass resolutions of 3.27×10^8 $h^{-1}\mathrm{M}_\odot$ (with the exception of Shin-Uchuu having an even higher resolution of 8.97×10^5 $h^{-1}\mathrm{M}_\odot$). These properties make Uchuu well-suitable for constructing high-accuracy galaxy and AGN catalogs, as sub-halo detection in dense environments is dependent on the resolution of the data (Comparat et al., 2020).

This work uses a mock halo catalog from the largest of the simulations in the suite, Uchuu, generated using the populating technique described in Comparat et al. (2020) and Comparat et al. (2025). Their study measures the cross-correlation between optical galaxy data and X-ray photons, in order to gain combined insight about galaxy distribution, the ICM and active galactic nuclei (AGN). Uchuu simulation data is populated with galaxies to obtain a mock catalog suitable for assessment of galaxy sampling uncertainties, more specifically to correct for systematic effects in the projected galaxy correlation function (Comparat et al., 2025). These qualities are specifically important also for substructure analysis.

The populating technique functions on the basis of X-ray emissivity profile construction from predictions of cluster emission. The profile modeling implemented in the method is empirical, allowing realistic cluster creation for meaningful performance

checks in this study, as well as modeled profiles being dependent only on halo mass, redshift and dynamical state parameters.

Before populating the DM only simulations with galaxies, they are processed with ROCKSTAR software (Behroozi et al., 2013) to obtain bound structures, that is, haloes and sub-haloes (Comparat et al., 2020; Ishiyama et al., 2021). The Uchuu simulation halo mass function at $z = 0$ overall agrees with the halo mass fitting function proposed by Despali et al. (2015) to difference of less than 10% for haloes with virial mass $M_{\rm vir} < 10^{15}h^{-1}{\rm M}_{\odot}$. The redshift range for the mock subcatalog used here is $0.20 < z < 0.22$.

In Comparat et al. (2020), snapshots are replicated and concatenated to create a full-sky shell. However, they discuss the limitations of this technique, such as the emergence of artefacts, particularly nearing the redshift range $z \sim 2$ as the number of replicas needed increases. This artefact issue is adressed by using the Uchuu simulation, as it mitigates the need for replications in lightcone generation due to its large volume. Consequently, the method can be used reliably to create the mock catalogue at a higher redshift range.

On the other hand, there is another issue to consider with simulations at the lower redshift range, namely noise at the high-mass end with limited volume as well as galaxy populations suffering from limited resolution. This problem is similarly avoided by Uchuu due to the large volume and high resolution that allow accurate cluster capture without significant noise issues.

The populating technique begins with coordinate definitions for the haloes. This is the most relevant stage of the method for this study, as the 3-dimensional group finding algorithms require positions in the plane-of-sky and redshift space. Coordinates in the mock catalog are defined by Comparat et al. (2020) as follows:

$$\text{R.A.} = \arctan 2(y, x)\frac{180}{\pi} + 180[^{\circ}], \tag{2.4}$$

$$\text{Dec.} = \arccos\left(z/d_C\right)\frac{180}{\pi} - 90[^{\circ}], \tag{2.5}$$

where (x, y, z) are the simulated positions of the haloes in the box, and $d_C = \sqrt{x^2 + y^2 + z^2}$[Mpc] is the comoving distance. The comoving distance – redshift relation then allows calculation of the real redshift z_R:

$$d_C(z) = \int_0^z \frac{c\ du}{H_0\sqrt{(1+u)^3\Omega_{m0} + \Omega_{\Lambda 0}}}[\text{Mpc}]. \tag{2.6}$$

Here H_0 is the Hubble constant, and Ω_{m0} and $\Omega_{\Lambda 0}$ are the matter and dark energy density parameters at the current time, respectively. The peculiar velocities of the haloes are then projected onto the line-of-sight using $v_{\parallel} = (v_x \cdot x + v_y \cdot y + v_z \cdot$

$z)/d_C$[km s^{-1}], so the peculiar motion-projected comoving distance can be calculated using the Hubble parameter, H, as

$$d_C^s = d_C + v_\parallel / H(z_R)[\text{Mpc}]. \tag{2.7}$$

Then, observed redshift z_S, i.e., the value needed for accurately replicating observed data, is calculated from d_C^s by again applying Equation 2.6.

As aforementioned, the complete method is, at its core, an X-ray cluster empirical model, aimed at accurately representing the clusters to be detected by the *eROSITA* X-ray telescope (Predehl et al., 2021). Briefly, the modeling process begins with surface brightness profile generation using existing observational samples from different surveys. Sets of simulated clusters characterized by predicted X-ray emissivity, redshift, temperature, and M_{500c}, are generated and linked back to the N-body simulation dark matter haloes with a 3-dimensional nearest neighbour procedure that accounts for the dynamical state of the clusters. For more detailed insights into the X-ray profile modeling steps of the procedure, the reader should consult Comparat et al. (2020).

In regard to substructure, Comparat et al. also repeat the above methods to generate subhaloes within massive haloes. The fraction of substructure generated depends on the mass resolution of the underlying simulation, making the choice directly correlated with the suitability of the resulting catalog to substructure analysis. The high resolution of the Uchuu simulations make them a well-reasoned choice for this method, resulting in the observed mock catalogue used in this work, consisting of 2524 galaxies and a total of 1372 subhaloes. It is worth pointing out that haloes can also be populated by only one galaxy, adding noise into the set as the population cannot be easily divided into subcomponents that members are divided amongst. Increased noise and complex subhalo distribution makes the dataset ideal for testing algorithm performance for substructure detection in challenging environments.

2.3 Group finding algorithms

Group finding algorithms are machine learning methods developed for classifying data into groups and determining distributions, among other statistical properties. For cluster cosmology, they are used for the purpose of identifying clusters or subcomponents from galaxy catalogs, and are built upon a variety of foundations and approaches. This subsection introduces four methods that all use different approaches for group member classification. It is also appropriate to note the distinction between group finders used to identify clusters, and utilizing them for substructure analysis within a large halo, for example. The approach taken in this thesis is the latter, which has received less

attention in previous literature.

A variety of other automated substructure analysis methods exist, such as the caustic technique by Yu et al. (2015), the mixture of Gaussian techniques by de Los Rios et al. (2016) and the group shape parameter-based identification method by Wen and Han (2013), to name a few (see also Tempel et al., 2017, and references therein).

2.3.1 Shifting gapper method

The first group finding algorithm to be explored for substructure and cluster merger analysis is the shifting gapper method described by Lopes et al. (2009). In their study, Lopes et al. use Sloan Digital Sky Survey (SDSS, York et al., 2000) data to estimate cluster properties for a supplemental catalog from Northern Sky Optical Cluster Survey (NoSOCS; Gal et al., 2000, 2003; Lopes et al., 2004; Gal et al., 2009). The shifting gapper method is applied particularly for interloper rejection in the process of estimating cluster velocity dispersion (σ_v), $R_{200/500}$, and $M_{200/500}$ for their chosen low-redshift SDSS subsample of $z \leq 0.1$.

Lopes et al. also apply pre-cleaning and preparation steps to their cluster data, such as removing duplicates, refining or re-determining photometric redshifts as well as sky positions, repeating data selection and using the updated data to determine richness, λ, and optical luminosity, $L_{\rm opt}$, estimates. The details of this preparation work are beyond the scope of this thesis and have not been carried out in the same manner for the optical datasets used here. It is nevertheless relevant to point out, as dataset statistics might affect method performance for a differing purpose and sample.

Before the virial analysis, Lopes et al. identify the spectroscopic redshift limits and select cluster members in the redshift space using the gap technique described in Katgert et al. (1996) and Olsen et al. (2005), but applying a density gap (Adami et al., 1998) instead of a fixed-value gap in the redshift distribution. This gap is dependent on the number of a cluster's galaxies in the redshift survey. Then, after a group significance assessment, they proceed with the "shifting gapper" technique for interloper rejection with a similar adaptation to Fadda et al. (1996).

This algorithm proceeds as follows: First, galaxies within a chosen aperture of 2.5 h^{-1} Mpc from the cluster center and a velocity difference limit of $|cz - cz_{\rm cluster}| \leq$ 4000 km s^{-1} are selected. Here z is the redshift of the galaxy, $z_{\rm cluster}$ is the redshift of the cluster, and c is the speed of light to convert redshift to velocity. Then, the maximum velocity offset, meaning the maximum out of $|v_{\rm low} - cz_{\rm cluster}|$ or $|v_{\rm high} - cz_{\rm cluster}|$, is determined and set as the symmetric velocity limit. Now, continuing to the actual shifting gapper procedure, galaxies within a chosen radial bin of 0.42 h^{-1} Mpc (0.60 Mpc for $h = 0.7$ assumed in this work) are selected and considered. If a radial offset of

$\geq 0.7h^{-1}$ Mpc (1 Mpc with $h = 0.7$) is found, all galaxies beyond this radial distance are discarded as interlopers. A larger bin is considered if less than 15 galaxies are found in the initial bin. Then, the gapper technique is run with a density gap size of $\Delta z = facgap\{1 + \exp[-(N-6)/33]\}$, where $facgap = \max\{|v_{low} - v_{high}|/10, 300\}$ and N is the number of galaxies. Identified groups are sorted in order of increasing offset from the center, and the closest one is chosen if the offset is within the limit $\leq c[0.001(1 + z_{\rm cluster})]$. In the case of no groups fulfilling this condition, all galaxies within $v_{\rm low}$ and $v_{\rm high}$ are kept and the algorithm proceeds to the next radial bin. If a valid group is found, $v_{\rm low}$ and $v_{\rm high}$ are updated according to the group, and they will be considered in the next radial bin instead.

Ultimately, the method was deemed not suitable for performing the subcomponent identification needed, due to filtering conditions too strict for the intended dataset of this study. Particularly, accuracy of the shifting gapper technique is expected to suffer from not meeting the outlined conditions, such as accurate photometric and spectroscopic redshift matching, where the following limit was set: $|z_{\rm photo} - z_{\rm spec}| < 0.03(1 + z_{\rm spec})$. Spectroscopic survey completeness limitations for unbiased σ_v, R_{200}, and M_{200} estimation are also discussed in Lopes et al. (2009), who find a survey completeness requirement of $M^* + 1$ for accurate virial analysis. This will constrain the applicability of the method to the intended dataset of this work with largely limited completeness.

2.3.2 Halo-based method

Another approach to group finding algorithms involves the dark matter (DM) haloes within whose potential the member galaxies reside. This subsection introduces the self-calibrating halo-based group finder described in Tinker (2021, 2022), which is based on the algorithm developed by Yang et al. (2005). The main feature of this method compared to previous halo-based techniques is the self-calibration, i.e. the method compares the catalog to complementary data, accounting for unknowns regarding the dichotomies between star-forming and quiescent galaxies and the connections between them and their haloes. Additionally, the model places emphasis on whether a galaxy is the central galaxy of a halo or a satellite orbiting within the host halo. This dichotomy is seen to have an effect on galaxy population properties (Tinker, 2021, and references therein).

One of the main ingredients for the algorithm is the total satellite luminosity $L_{\rm sat}$, and the luminosity of the central galaxy $L_{\rm cen}$. These are separated for star-forming vs. quiescent central galaxies, and the group finder is fit to relative measurements of $L_{\rm cen}$ and the ratio $L^{\rm q}_{\rm sat}/L^{\rm sf}_{\rm sat}$. The galaxies themselves are separated into star-forming and

quiescent using D_{4000}, the ratio of the average flux densities above and below the 4000 Å break.

The algorithm makes classifications using probabilities: it determines the probability that neighbouring galaxies are satellites within the halo of a central galaxy. This also involves an initial estimate of the halo mass. Tinker sets the probability requirement $P_{\rm sat} > 0.5$ for a neighbour to be accepted as a satellite, where

$$P_{\rm sat} = [1 - (1 + P_{\rm proj} P_{\rm z} / B_{\rm sat})^{-1}]. \tag{2.8}$$

Here, $P_{\rm proj}$ and $P_{\rm z}$ are the projected and line-of-sight probabilities, respectively, and $B_{\rm sat}$ is a free parameter that represents the threshold probability describing the likelihood of the satellite being a member of a larger halo, and is a function of $L_{\rm gal}$.

Altogether, including the thresholds and weighting factors, the algorithm has 14 free parameters. Running the algorithm begins by setting starting values for all free parameters. The group finder is then run with these values and iterates until the fraction of determined satellites begins to converge.

The group finder initally labels all galaxies as central and sorts them by decreasing halo mass. Then, beginning with the largest halo, the probabilities of the neighbouring galaxies being members are calculated (i.e., $B_{\rm sat}$). If the threshold is met, the galaxy is re-classified. The total luminosity for each group is then summed, and groups are again sorted by descending luminosity. After assignment of members, inverse abundance matching is used to assign halo masses to the groups based on the host halo mass function, taking into account group property-based weighing (i.e. star-forming or quiescent central galaxies). The output of the group finder is then a list of groups with assigned halo mass and each assigned member galaxy classified as central or satellite.

This is followed by comparing group catalog predictions to observational data. Predictions are made using the Bolshoi-Planck N-body simulation MultiDark (Klypin et al., 2016), which produces a group catalog to populate haloes from a given set of parameters. The simulation is run with the halo-occupation distribution (HOD; Wechsler and Tinker, 2018) from the group catalog obtained from the group finder. Tabulated values for satellite luminosities as a function of halo mass M_h are then used to estimate $L^{\rm q}_{\rm sat}/L^{\rm sf}_{\rm sat}$. Finally, the goodness-of-fit is calculated by comparing the group catalog predictions to the observational data.

The halo-based method also involves challenges for this study. It is more complex, as it considers the ratio of projected and line-of-sight probabilities and involves guesses for halo mass from previous data, making applications for new datasets possibly limited. Tinker (2021) also define haloes as distinct and not within the radii of larger haloes, which plays a role in the performance on substructure analysis in particular. This could potentially mean significant re-training for use in intra-cluster analysis. On the other

hand, probabilistic attachment of galaxies to the groups is relevant. Group finding using Tinker method has been applied by Damsted et al. (2023), though in their study the method was used to clean projected components.

2.3.3 Friends-of-Friends algorithm

Friends-of-friends (FoF) algorithms are a type of group finder that uses (usually user-defined) distance thresholds, called linking lengths, to determine whether two points are connected to each other and belong to a group. Directly connected points, in this case galaxies, are considered "friends" if the distance between them falls within the defined linking length. A galaxy is then assigned to a group or cluster with its friends, as well as the friends of these galaxies, "friends of friends".

The specific FoF algorithm studied in this work is described in Tempel et al. (2016). This algorithm functions in 3-dimensional space, namely two spatial and one velocity dimension. For astronomical uses, linking length should additionally scale with distance. Tempel et al. (2017) use the following linking length definition as a function of z:

$$d_{\mathrm{LL}}(z) = d_{\mathrm{LL},0}[1 + a \arctan(z/z_*)], \tag{2.9}$$

where $d_{\mathrm{LL},0}$ is the linking length at $z = 0$, and a and z_* are both free parameters. Additionally, a ratio between the radial line-of-sight and transversal skyplane linking lengths ($b_\parallel$ and $b_\perp$, respectively) is set to define a cylindrical volume. The value chosen by Tempel et al. (2017) in their SDSS study is $b_\parallel/b_\perp = 12$, and the same value was used on the run with Uchuu simulation data in this work.

Conventional FoF group finders are widely and successfully used for cluster identification, but have known limitations and are susceptible to overmerging (Knobel et al., 2009). This could result in merger event substructure remaining undetected, as well as field galaxies or filaments being absorbed into the detected cluster (Tempel et al., 2016). Consequently, Tempel et al. have implemented an additional refinement step using model-based clustering with `mclust` (see Section 2.3.4). Overmerging was also observed with the Uchuu data using the FoF galaxy group finder. The issue persisted with the further refinement using `mclust`, while improvements were seen when using `mclust` only for substructure detection.

2.3.4 *mclust:* Gaussian mixture modelling

Alternatively to using Gaussian mixture modelling as part of a refinement procedure as is done in Tempel et al. (2016), model-based clustering methods also form a discrete

class of group finders. Due to the overmerging issues described in Section 2.3.3, the possibility of using `mclust` directly for substructure analysis was explored. This section discusses the use and functionality of `mclust` (Scrucca et al., 2023), closely following descriptions provided in Scrucca et al. (2016).

The `mclust` algorithm is a Gaussian mixture modeling algorithm (GMM). At their core, mixture models are linear combinations of models that describe components of a dataset to make up the total distribution. Gaussian mixture models apply multivariate normal distributions to these subcomponents.

Let $\boldsymbol{x} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, ..., \boldsymbol{x}_n\}$ be a sample of identically distributed observations, for which a probability density function can be formulated through a finite mixture model of G components:

$$f(\boldsymbol{x}_i; \Psi) = \sum_{k=1}^{G} \pi_k f_k(\boldsymbol{x}_i; \boldsymbol{\theta}_k), \tag{2.10}$$

where G is the number of mixture components, in this case the number of clusters in the data, $\Psi = \{\pi_1, ..., \pi_{G-1}, \boldsymbol{\theta}_1, ..., \boldsymbol{\theta}_g\}$ are the parameters of the model, $(\pi_1, ..., \pi_{G-1})$ are the mixing probabilities, and $f_k(\boldsymbol{x}_i; \boldsymbol{\theta}_k)$ is the kth component density for $\boldsymbol{x}_i$, the observation, with a parameter vector $\boldsymbol{\theta}_k$.

For Gaussian mixture models, a multivariate Gaussian distribution is assumed for each component, meaning $f_k(\boldsymbol{x}; \boldsymbol{\theta}_k) \sim N(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$, where $\boldsymbol{\mu}_k$ is the mean vector at which clusters are centered at, and $\boldsymbol{\Sigma}_k$ is the covariance matrix. The covariance matrix determines other features, such as shape and volume, and is given by

$$\boldsymbol{\Sigma}_k = \lambda_k \boldsymbol{D}_k \boldsymbol{A}_k \boldsymbol{D}_k^T, \tag{2.11}$$

where $\boldsymbol{A}_k$ controls the shape of the density contours, $\boldsymbol{D}_k$ determines their orientation, and λ_k controls the volume.

The prodecure involves determining the mixture model parameters, Ψ, via the maximum likelihood estimator (MLE), as maximizing the full log-likelihood function is often too resource-intensive. Obtaining the MLE is done using an expectation-maximization (EM) algorithm, which is implemented into the `mclust` package. The Bayesian Information Criterion (BIC) is utilized to determine the number of mixing components, G, and the covariance parametrization.

The different available models vary in which components are equal or variable, namely the volume, shape and orientation. In other words, the distribution is allowed to be spherical, diagonal or ellipsoidal, which consequently means different expressions for the covariance matrix $\boldsymbol{\Sigma}_k$. Ver. 5 of `mclust` provides 14 different models with varying geometric characteristics, summarized visually in Figure 2.6.

Model `VVV` was chosen for galaxy clustering analysis because it describes the most

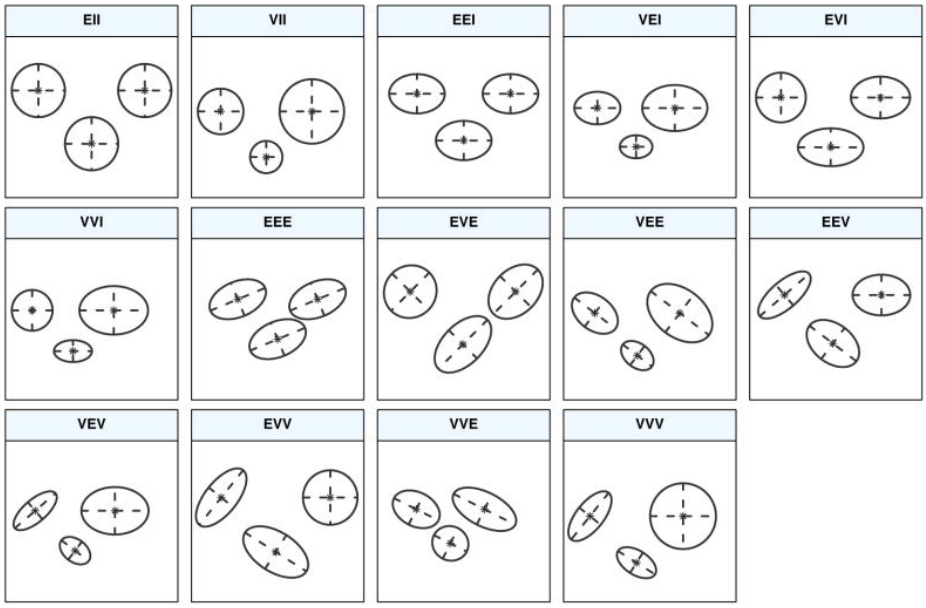


Figure 2.6: Visualization summarizing isodensity ellipses with varying (un)fixed components for each model available in `mclust` V5. From Scrucca, L., Fop, M., Murphy, T. B., & Raftery, A. E. (2016). mclust 5: Clustering, Classification and Density Estimation Using Gaussian Finite Mixture Models. *The R Journal*, *8*(1), 289–317. https://doi.org/10.32614/RJ-2016-021 .

realistic results in the context of member galaxy distribution within clusters by allowing varying shape, volume, and orientation of the components.

2.4 Cluster Validity Indexes

The group finders used for galaxy cluster and subcluster detection are generally unsupervised clustering algorithms. This means that they run without class-labelled training data as input along with the data to be classified. This has apparent advantages due to no need for ground-truth comparison data, and hence applicability to a wide variety of datasets without significant re-training. However, one significant drawback involves the complexity of clustering validity evaluation, i.e. the assessment of whether clustering results are acceptable (Tang, 2022, p. 1-22).

Results are most commonly evaluated in terms of metrics that describe the tightness of detected groups, as well as separations between groups. These metrics, called Cluster Validity Indexes (CVIs), are dependent on the specific dataset in use, and can hence be an incomplete description of the solution. Therefore, diverse evaluation metrics founded on multiple solution characteristics are necessary to perform thorough assessment. This section describes five metrics used in this work to evaluate substructure detection results, comparing the aforementioned qualities as well as purity and possible randomness of assignments. Table 2.1 summarizes the main properties of the

Table 2.1: Summary of metrics described in Sections 2.4.1-2.4.5.

Metric	Type	Concept	Range	Interpretation
ARI (Hubert & Arabie, 1985)	External	Randomness-based agreement between partitions	$[-0.5; 1]$	The closer to 1, the better
V-measure (Rosenberg & Hirschberg, 2007)	External	Harmonic mean of purity and homogeneity	$[0; 1]$	The closer to 1, the better
Silhouette (Rousseeuw, 1987)	Internal	Self-similarity of clusters compared to neighbours	$[-1; 1]$	The closer to 1, the better
Calinski-Harabasz (Caliński & Harabasz, 1974)	Internal	Ratio of between-cluster and within-cluster dispersion	$[0; \infty)$	The higher, the better
Davies-Bouldin (Davies & Bouldin, 1979)	Internal	Average worst scatter and distance ratio for cluster pairs	$[0; \infty)$	The lower, the better

metrics, their ranges, and interpretations.

2.4.1 Adjusted Rand Index

The Rand Index (RI; Rand, 1971) is a measure of agreement between two clustering solutions, or partitions, making it an intuitive choice for assessing how well clustering algorithms recover known ground-truth solutions when they are available. It describes how likely it is for a randomly chosen pair of data points to be in agreement between the two partitions, i.e. whether two points are in the same or different clusters in both partitions. In a simplified form, the index is given by

$$\mathrm{RI} = \frac{A}{\binom{n}{2}}, \tag{2.12}$$

where A is the total number of agreements. In more detail, given a set of objects $S = \{o_1, ..., o_n\}$ and two partitions of S, $U = \{u_1, ...u_R\}$ and $V = \{v_1, ...v_C\}$, where objects can only be members of one cluster in a given partition, an "agreement" is a case where a randomly chosen pair of objects in S are either in the same group in U and in the same group in V, or in different groups in both partitions. However, varying definitions of agreement can be applied, for example considering only the first condition as an agreement.

Later, a correction to the Rand Index was added by Hubert and Arabie (1985) in order to account for chance and to normalize the indices under a chosen "null model", in this case referring to a purely random clustering. This is done by applying an expected value for the RI, supplying the index with a baseline for comparison. The resulting corrected index is referred to as the Adjusted Rand Index (ARI) and takes the following form:

$$\mathrm{ARI} = \frac{\mathrm{RI} - E(\mathrm{RI})}{MAX(\mathrm{RI}) - E(\mathrm{RI})}. \tag{2.13}$$

The ARI takes values close to 0 when the index is equal to the expected value (i.e. labeling is random) and 1 for perfect matching between U and V. Negative values are allowed, corresponding to clustering in worse than random agreement. In such a case, a completely random classification of points would yield a higher number of correct assignments than the partition in comparison.

It is worth pointing out that the ARI is, for the purposes of this work, an external evaluation metric, meaning that it employs information not inherent to the dataset and clustering results, in this case the class labels.

2.4.2 V-measure

The validity measure known as V-measure (Rosenberg & Hirschberg, 2007) is another external measure for clustering evaluation, describing the homogeneity and purity* of a partition. Specifically, the V-measure is the harmonic mean between clustering homogeneity and purity. Homogeneity describes the class similarity within groups in the partition, meaning that a perfectly homogenous cluster would only have members with the same class label. Rosenberg and Hirschberg describe this as "zero entropy" within the cluster. Purity is defined similarly, but approaching from the class labels, and so it describes how well objects with the same class label have been assigned to a single cluster by the procedure. The full expressions for calculating the homogeneity and purity are beyond the purpose of this section, see Eq. 1 and 2 in Rosenberg and Hirschberg (2007) for more detail.

The weighted harmonic mean of homogeneity and purity is then

$$V_\beta = \frac{(1+\beta) \times h \times c}{(\beta \times h) + c}, \tag{2.14}$$

where h and c are the homogeneity and purity, respectively, and β is the weight, controlling whether c or h is weighted more strongly. If $\beta > 1$, purity is more strongly weighted, and if $\beta < 1$, homogeneity is. $\beta = 1$ is applied in this work. For perfectly matching partitions, $c = h = 1$, and $V_\beta = 1$. The lower bound of the score is zero.

An important feature of the V-measure is that it is independent of the number of classes and clusters, as well as the size of the dataset, meaning that the value can be computed for any clustering solution.

*The term "purity" is typically replaced by "completeness" in the definition and discussions of the V-measure. However, the former term is implemented here to avoid confusion with the spectroscopic completeness of a sample with respect to the cluster population, elsewhere in the work also referred to as just completeness.

2.4.3 Silhouette

The Silhouette, introduced by Rousseeuw (1987), is a measure of the similarity of points with their assigned group compared with another. It is hence a measure of cohesion within the partition, describing the tightness and separation of clusters. An important benefit of the Silhouette score is that it is an internal metric, and can hence be used to characterize clustering solutions' success without the need of correct class label availability.

In calculating the Silhouette score, similarity is based on distances between points and the rest of their cluster members. In particular, for each object i, an average dissimilarity $a(i)$ within assigned cluster A with n_A members is computed by

$$a(i) = \frac{1}{n_A} \sum_{i=1, i \neq j}^{n} d(i, j), \tag{2.15}$$

where $d(i, j)$ is the distance between objects i and j. The distance from the object to itself, $d(i, i)$, is intuitively not included in the sum. Similarly, the average dissimilarity of i is also calculated with respect to any other cluster C, denoted by $c(i, C)$. Now, the neighbour B of cluster A is defined by $b(i) = \min(c(i, C))$, that is $b(i) = c(i, B)$. The neighbour can be understood as the "next-best-option" for classifying object i.

Finally, the Silhouette score $s(i)$ can be calculated using the average dissimilarities of i compared to its own cluster A and neighbour B:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}. \tag{2.16}$$

Values of the Silhouette score then lie in the range $-1 \leq s(i) \leq 1$, where a value of 1 refers to very distinct clusters and high certainty that i has been assigned to the best-matching cluster, while values close to -1 indicate that a significantly better classification was available but not assigned. The mean Silhouette score over the entire dataset is hence a signifier of how tight and distinct clusters are in a given solution. Notably, the Silhouette value is generally strongly affected by the dataset used, and can have low scores regardless of the solution.

2.4.4 Calinski-Harabasz Index

The Calinski-Harabasz index (CH; Caliński and Harabasz, 1974) or Variance Ratio Criterion compares the overall separations between clusters to the dispersion of a cluster. It is an internal metric, and, similarly to the Silhouette, is a descriptor of how distinct clusters are in a given partition.

Cluster separations are used to calculate the between-cluster dispersion matrix ($BCSS$), which is the distance between each cluster's centroid c_i to the general cen-

troid c, both of which are calculated within the procedure, weighted by the number of members in the cluster n_i, summed over all clusters k:

$$BCSS = \sum_{i=1}^{k} n_i \times |c_i - c|^2. \tag{2.17}$$

Comparison within individual clusters is done by computing the within-cluster dispersion matrix, the sum of the distance between each object x in a cluster and the cluster's centroid c_i, and again summed over all clusters in the partition:

$$WCSS = \sum_{i=1}^{k} \sum_{x \in C_i} |x - c_i|^2. \tag{2.18}$$

The Calinski-Harabasz index is defined as the ratio of the between-cluster and within-cluster matrices, normalized by the degrees of freedom in order to minimize bias resulting from the magnitude of k. Denoting the total number of objects in the set as n, the CH index is calculated from

$$CH = \frac{BCSS/(k-1)}{WCSS/(n-k)}. \tag{2.19}$$

Resulting values have a lower bound of zero, and there is no upper bound. The higher the value, the larger the difference between between-cluster and within-cluster dispersions, which signifies tighter and more distinct clusters.

2.4.5 Davies-Bouldin Index

The Davies-Bouldin index (DBI; Davies and Bouldin, 1979) can, in general terms, be understood as a similar measure to the Calinski-Harabasz index, as it also compares the scatters within clusters to the between-cluster distances. However, the Davies-Bouldin index, along with the Silhouette, have been suggested to to be more reliable compared to the Calinski-Harabasz index, particularly in the context of convex-shaped clusters (Chicco et al., 2025).

To calculate the DBI, a cluster similarity measure R_{ij} is first defined. This measure is defined from the ratio between the scatter S_i within a cluster C_i with n_i members, and the between-cluster distance M_{ij}, where j denotes another cluster satisfying $i \neq j$. Again denoting object positions by x and the cluster centroid by c_i, the scatter and distances are defined as

$$S_i = \frac{1}{n_i} \sum_{x \in C_i} |x - c_i| \tag{2.20}$$

and

$$M_{ij} = |c_j - c_i|. \tag{2.21}$$

Then, the similarity measure R_{ij} is calculated from the sum of the scatters for C_i and C_j and the distance between them:

$$R_{ij} = \frac{S_i + S_j}{M_{ij}} \tag{2.22}$$

Finally, the DBI is calculated from the maximum values of the similarity measure for each cluster, calling $R_i = \max_{j \neq i} R_{ij}$ and again denoting the total number of clusters by k:

$$\overline{R} = \frac{1}{k} \sum_{i=1}^{k} R_i \tag{2.23}$$

The DBI hence describes the average worst (largest) ratio between cluster pairs' scatter and distance with respect to each other, over the whole partition. Outputs then range from zero to infinity, where smaller values indicate tight and distinct clustering solutions.

3. Results and analysis

3.1 Substructure detection from simulation

Running the `mclust` algorithm on the whole Uchuu subset using the plane-of-sky positions and observed redshifts as input yields in the detection of nine subgroups within the set. No priors are supplied to the algorithm, and `mclust` is run with default input variables. With these settings, arrangements with a number of mixture components within the range $G = 1 : 9$ are tested. Figure 3.1 outlines the clustering results from each plane alongside the halo information from the simulation catalog. Specifically, comparisons are made with respect to the largest subhaloes that galaxies are members of, as the halo catalog supplies a hierarchy of subhalo information.

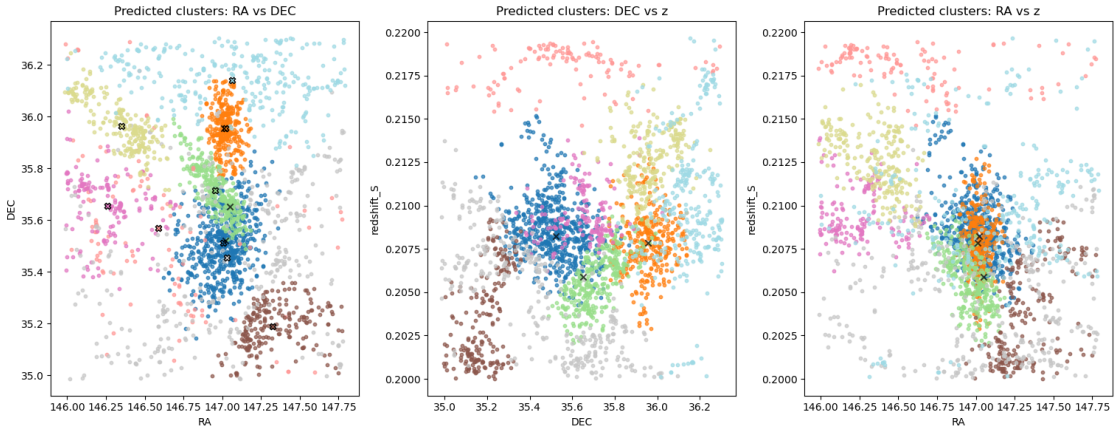
Raw clustering results on the complete set recreate the general subcomponents of the dense central main component, but significant overmerging and noise is apparent within the sparser edge regions with less rich haloes. As one of the main aims of this work is to emprically determine important completeness thresholds for substructure detection, the performance of `mclust` is studied further at reduced completeness levels in the range $10\% < c < 100\%$.

This was first approached by randomly removing varying fractions of redshift data from the mock set. Figure 3.2 shows results of one `mclust` run when redshift completeness has been reduced to 85% and 65% completeness. These values have been chosen to correspond to the completeness thresholds reported for the Low-z bright (84.1%) and Low-z faint (65%) subsurveys in the CHANCES 4MOST Low-z survey in Méndez-Hernández et al. (2026).

Subcomponent detection is then repeated in the low-completeness regime. This is specifically of interest in this study due to spectroscopic survey limitations (for example, data availability as of DESI Data Release 1, DESI Collaboration et al., 2026), which often lead to spectroscopic cluster data availability extending from 50% to even 10% completeness (see for example SPIDERS; Clerc et al., 2016; Damsted et al., 2024).

Analysis with `mclust` is hence also repeated for 50%, 20%, 15%, and 10% completeness, and results of one run are shown in Figure 3.3 as an example. Notably, the overall central main component is still detected as a distrinct structure at 10%

(a) `mclust`



(b) Uchuu halos

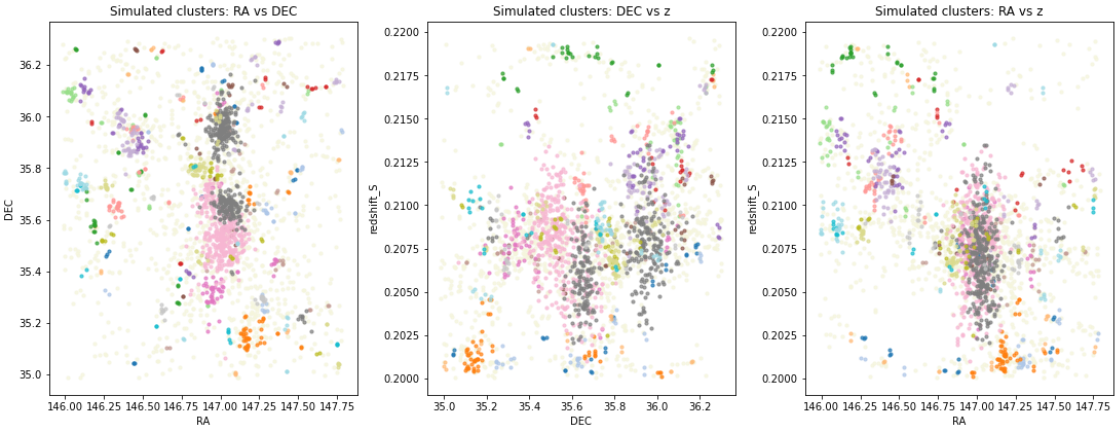


Figure 3.1: Substructure detection results from `mclust` applied to the whole Uchuu subset are shown in (a). Nine subcomponents are detected, where galaxies assigned to the same component are marked with the same color. Figure (b) presents the halo catalog supplied in the dataset, where galaxies belonging to no halo with other galaxies are marked in light beige. From left to right, figures show results in coordinates ra [°] vs. dec [°], dec vs. z_S, and ra vs. z_S, where z_S is the observed redshift. Coloured crosses denote the plane-of-sky cluster centers by average coordinates of the subcluster with matching colour, and non-coloured crosses show the cluster centers of the three largest haloes in the sample, given in the simulation catalog.

completeness, though the three central subcomponents are no longer resolved from one another below 20%.

3.2 Evaluation of metrics

The performance of `mclust` is assessed by computing the Adjusted Random Index, V-measure, Silhouette, Calinski-Harabasz, and Davies-Bouldin metrics using the measures described in Section 2.4. Briefly, the external metrics ARI and V-measure describe how well the clustering solution agrees with the truth labels given, i.e. the simulation membership catalog, while internal metrics describe properties of the partition itself.

Computations and analysis are limited to subhaloes of known richness $\lambda > 30$, in order to place focus predominantly on the richest subcomponents due to their significance in recovering cluster dynamical properties. With the limited number of rich subhaloes in the dataset, taking a completeness-reduced subsample and running group finding with `mclust` is repeated ten times at each completeness level. The five metrics are then calculated using built-in functionalities in the `scikit-learn` Python package (Pedregosa et al., 2011). Table 3.1 summarizes the obtained results.

In order to better evaluate the competitiveness of spectroscopic substructure detection with `mclust`, results are compared to substructure clustering results from photometry, presented by Sepúlveda et al. (in prep.). They study the performance of nine algorithms on three different simulated photometric datasets by calculating the same metrics used in this work. Two of the used algorithms are chosen for comparison: Gaussian Mixture Modeling and Density-Based Spatial Clustering of Applications with Noise (DBSCAN; Ester et al., 1996; Schubert et al., 2017).

Furthermore, out of the three datasets defined by Sepúlveda et al., the results obtained from their dataset S2 are chosen for comparison with spectroscopy due to a similar application of richness cuts and hence closest representation of the mock dataset used here. More specifically, Sepúlveda et al. define the dataset S2 as the "clean set", where only subhaloes selected as substructures are included in analysis. They have chosen to consider subhaloes with above 4 galaxies as substructures, hence still including smaller structures compared to the current simulation set and cuts applied. Metric values reported by Sepúlveda et al. for both GMM and DBSCAN on the set S2 are also presented in Table 3.1. Despite the outlined differences, and hence limitations imposed on direct comparison of clustering solutions, the study was chosen due to the scarcity of similar statistical studies of group finders for substructure detection.

Because internal metrics are highly dependent on the underlying properties of the dataset itself, it is appropriate to compute the metrics directly on the truth-labelled dataset as well, namely the simulation catalog. All metrics are hence also computed

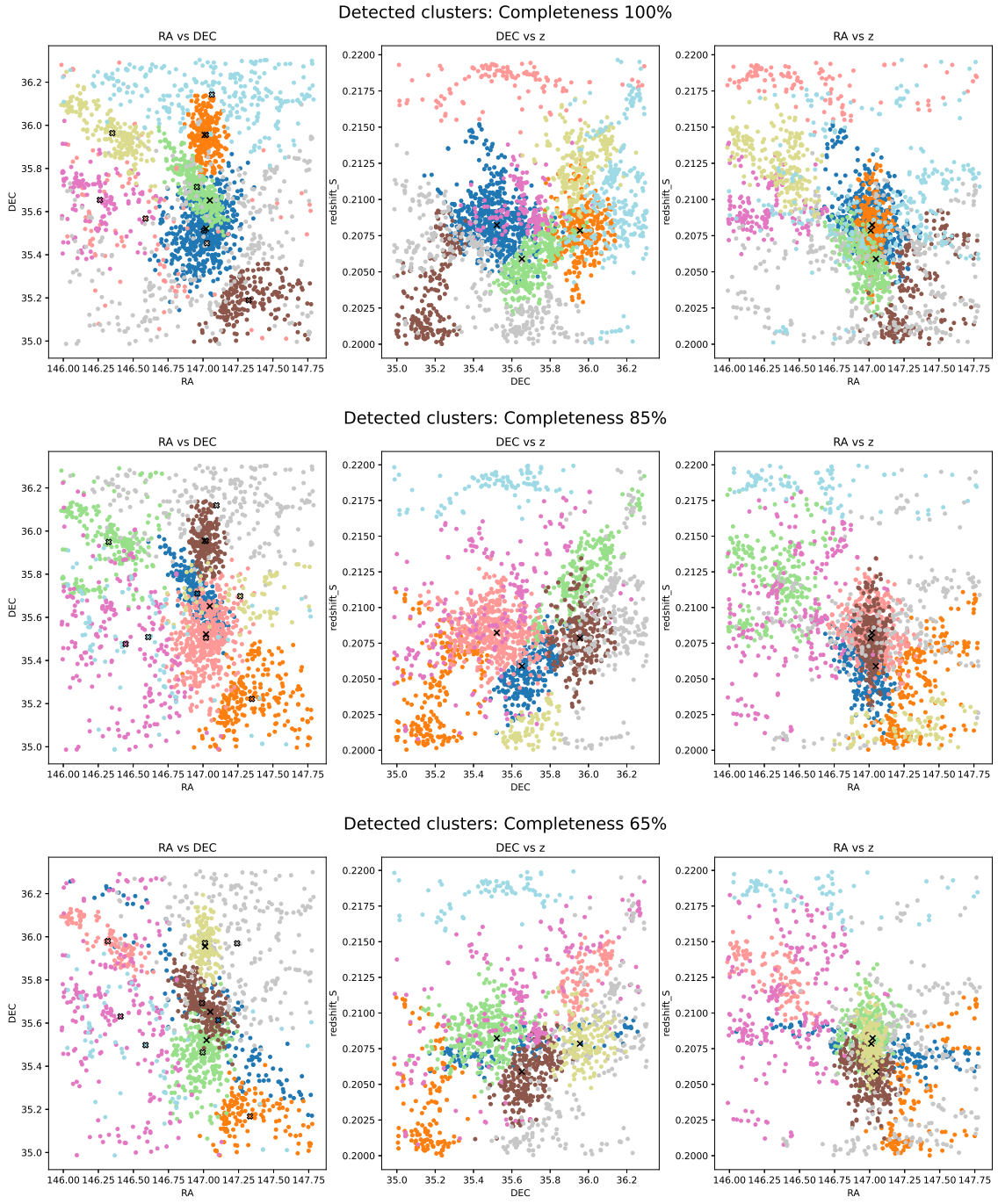


Figure 3.2: Clusters detected with `mclust` at completeness levels 100%, 85%, and 65%, from top to bottom. Subfigure axes and cluster centers as described in Figure 3.1.

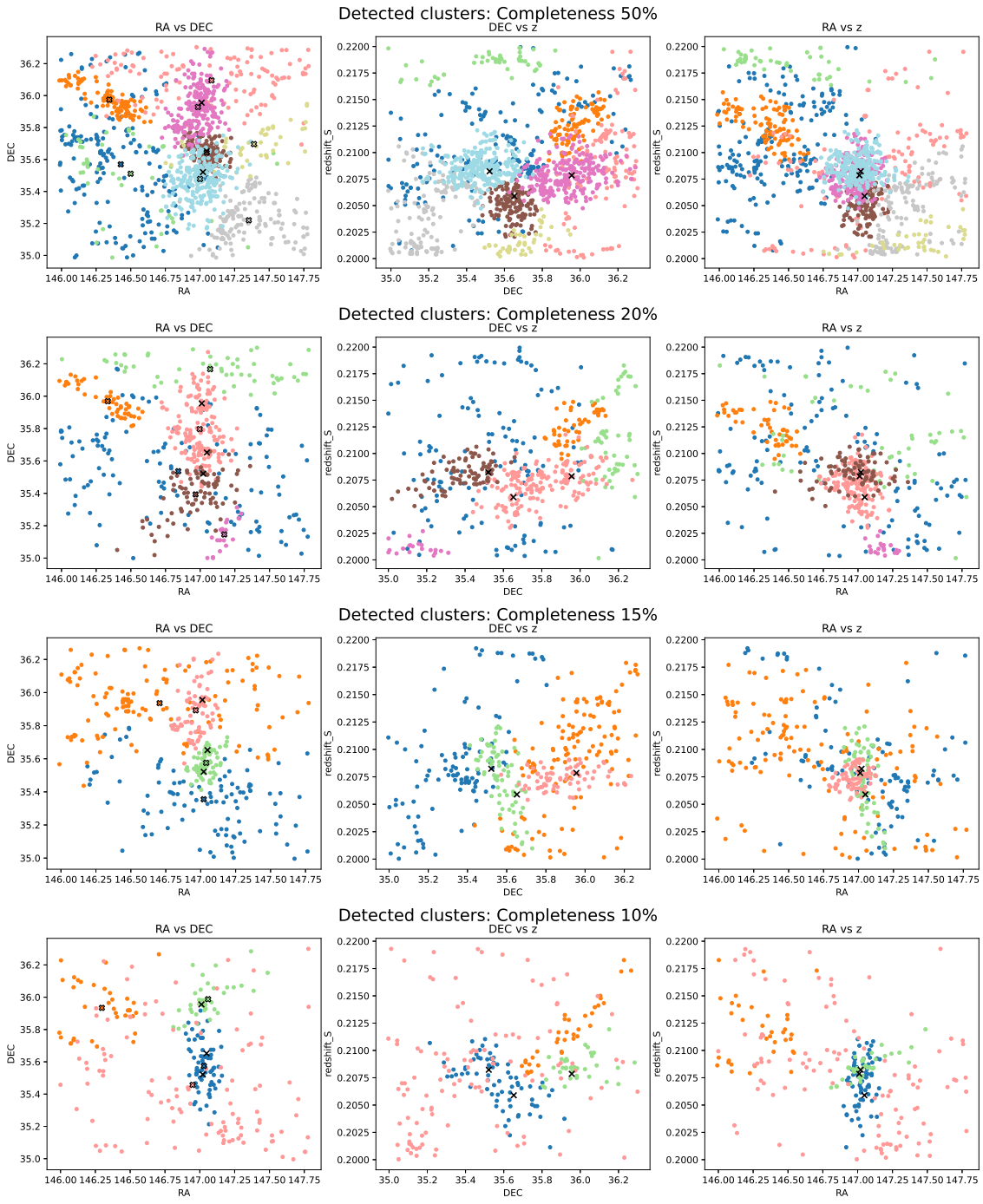


Figure 3.3: Clusters detected with `mclust` at completeness levels 50%, 20%, 15%, and 10%, from top to bottom. Subfigure axes and cluster centers as described in Figure 3.1.

Table 3.1: Evaluated mean metrics for the `mclust` clustering solution, calculated considering only substructures with $\lambda > 30$. Metrics are calculated for completeness values 10%, 15%, 20%, 50%, 65%, 85%, and 100%, across ten samples at each level. The results from Sepúlveda et al. (in prep.) on their dataset S2 with GMM and DBSCAN, as well as values calculated directly from the truth-labelled mock catalog are shown on the bottom rows as reference.

Completeness (%)	ARI	V-measure	Silhouette	Calinski-Harabasz	Davies-Bouldin
10	0.039 ± 0.027	0.103 ± 0.030	-0.053 ± 0.053	2.541 ± 0.878	4.466 ± 1.081
15	0.327 ± 0.081	0.399 ± 0.076	0.054 ± 0.079	60.898 ± 21.803	5.064 ± 1.929
20	0.541 ± 0.038	0.594 ± 0.026	0.241 ± 0.045	134.708 ± 14.848	1.519 ± 0.230
50	0.569 ± 0.035	0.597 ± 0.016	0.100 ± 0.041	271.600 ± 17.352	2.121 ± 0.268
65	0.631 ± 0.007	0.624 ± 0.009	0.187 ± 0.035	399.188 ± 29.405	1.967 ± 0.457
85	0.598 ± 0.013	0.598 ± 0.005	0.144 ± 0.036	422.050 ± 16.727	2.119 ± 0.349
100	0.633 ± 0.006	0.621 ± 0.006	0.037 ± 0.042	501.200 ± 21.694	1.585 ± 0.203
GMM Sepúlveda (S2)	0.186 ± 0.189	0.303 ± 0.178	0.376 ± 0.196	96.908 ± 122.583	0.609 ± 0.061
DBSCAN Sepúlveda (S2)	0.151 ± 0.093	0.329 ± 0.151	0.091 ± 0.103	11.925 ± 9.516	3.509 ± 3.116
Truth labels (100%)	1.0	1.0	0.402	1065.883	0.769

for the known haloes at 100% completeness with the same richness cut of $\lambda > 30$, and values are added in Table 3.1. The internal metrics can then be thought of as reference values; a good clustering solution for the present dataset should approach these values, rather than the general ideal values summarized for example in Table 2.1.

Both external metrics, ARI and V-measure, increase with increasing completeness, but plateau to ~ 0.6 above completeness $\sim 50\%$. While there is improvement in the ARI from completeness 50% to 65%, the increase from 65% to 85% no longer improves the result. The ARI surpasses results from Sepúlveda et al. for both GMM and DBSCAN at completeness 15%, meaning that at and above 15% spectroscopic completeness, the `mclust` clustering solution is less likely to be randomly assigned than the partitions obtained from the photometric dataset. The V-measure shows similar behaviour, signifying `mclust` outperforming the photometric runs in terms of homogeneity and purity at the same level of 15% spectroscopic completeness.

Comparing the self-similarity of detected groups with the Silhouette metric does not indicate similarly clear outperformance over photometry. Improvements over DBSCAN from Sepúlveda et al. are seen at 20% completeness, but `mclust` does not outperform GMM for the photometric data at any of the completeness levels. However, direct comparison is challenging, as truth label metrics are not available for the dataset S2 in Sepúlveda et al. (in prep.). Yet, comparison of how well the solutions approach

the truth label values is more statistically meaningful than comparing the metrics on different datasets directly. The Silhouette value of the truth labels on the mock subset is $s = 0.402$, highlighting that substructures are not significantly distinct from one another and the dataset is noisy. It is worth pointing out that the Silhouette metric drops at 100% completeness, likely as a result of the increasing density and limited number of mixture components used.

The Calinski-Harabasz index increases monotonically throughout the completeness levels. While the CH index also describes how distinct groups are, it compares the dispersion within clusters to dispersion between clusters, whereas the Silhouette can be understood as describing how well points could have been assigned to the "next-best option". The monotonic growth hence suggests that `mclust` consistently selects denser regions as subcomponents and distinguishes them from the sparser regions in-between. The value for the truth labels is CH = 1065.883, and the maximum CH value obtained with `mclust` is CH $= 501.2 \pm 21.7$ at completeness 100%. Similarly to what is seen for the external metrics, the CH value indicates improvements on the solution already at low completenesses compared to photometry: GMM at CH $= 96.908 \pm 122.583$ is overtaken at 20% completeness, while DBSCAN performs worse than `mclust` at 15% completeness. The large uncertainties of both the GMM and DBSCAN results also indicate significantly higher scatter in the dispersions between clusters, which is not seen in spectroscopic subcomponents with `mclust`, which have lower relative errors across all completeness levels. However, comparison with the truth label results on S2 would be necessary to understand the effects of underlying properties of the different simulated datasets.

Finally, the Davies-Bouldin index describing the average ratio between the largest scatter and between-cluster distance is investigated. Remembering that values closer to zero indicate more distinguishable groups, results are seen to improve at completeness 20%, but do not improve further with increasing completeness. The GMM DBI value of DBI $= 0.609 \pm 0.061$ on photometry is lower than results on the spectroscopic dataset at all completenesses. DBSCAN yielded the value DB $= 3.509 \pm 3.116$ on the S2 set, which is higher than the spectroscopic set already at 20% completeness. However, the error range is also much higher for this value than for GMM, suggesting that scatter for this method can be higher than for the two Gaussian Mixture Models. When instead comparing to the truth label value of DBI = 0.769, which is greater than the result for GMM on S2, it is apparent that the scatter of the dataset itself sets a lower bound for the DB index, in which case the results set S2 cannot be directly compared for the Davies-Bouldin index. Nevertheless, it is seen that Gaussian Mixture models compete with and outperform the density-based clustering method DBSCAN given these metrics, while some inconsistency remains between overall comparison between

spectroscopy and photometry.

3.3 Membership matching

External evaluation metrics, notably the V-measure used here, involve matching the true label catalog with the detected group catalog in order to describe how successfully the solution has been recovered. As described in Section 2.4.2, the distribution of members into detected groups is calculated by computing the homogeneity (approaching from the detected groups) and purity (approaching from the truth labels). Effectively, calculation of the purity also reveals the best-matching detected group corresponding to each true component, which in itself can be used to characterize algorithm performance and is hence further investigated in this section.

Thorough catalog matching involves various characteristics and functionalities. In this instance, where the simulation catalog is available to be used as the reference class labels, membership matching is done from the true haloes to the detected groups. Bidirectional cross-matching can also be applied, and is available for example in the catalog matching framework developed for LSST data, ClEvaR ("ClEvaR (0.13.2)", 2021). Such matching is particularly useful for comparisons of observational catalogs.

The matching procedure formulated here also follows a similar logic as laid out in the membership-based catalog matching functionality in ClEvaR. However, haloes are filtered by richness ($\lambda > 30$) before matching, as described in the previous section. Then, fractions of how members were classified into different groups by `mclust` are computed. The largest fraction is named the best fractional match, and the corresponding group is assigned as the matching pair. As aforementioned, the magnitude of the fractional match is then a measure of the purity of the assignment.

To study how the membership matching evolves with reduced completeness, ten samples are drawn from the dataset for each completeness threshold (10%, 15%, 20%, 50%, 65%, 85%, and 100%) in order to obtain a larger total number of components viable for analysis. The best fractional matches are plotted against completeness in Figure 3.4. Matches are above 80% throughout the whole data completeness range, suggesting high purity in the clustering solution for rich substructures. A negative trend is observed in the matching, which is expected as the number of members in groups as well as the number of detected components increases; the increasing density in the central region increases the likelihood of members being classified into other subgroups.

The most notable result for fractional matching is seen at completeness $\gtrsim 50\%$, beyond which data completeness does not significantly affect the best fractional match, i.e. the assignment completeness.

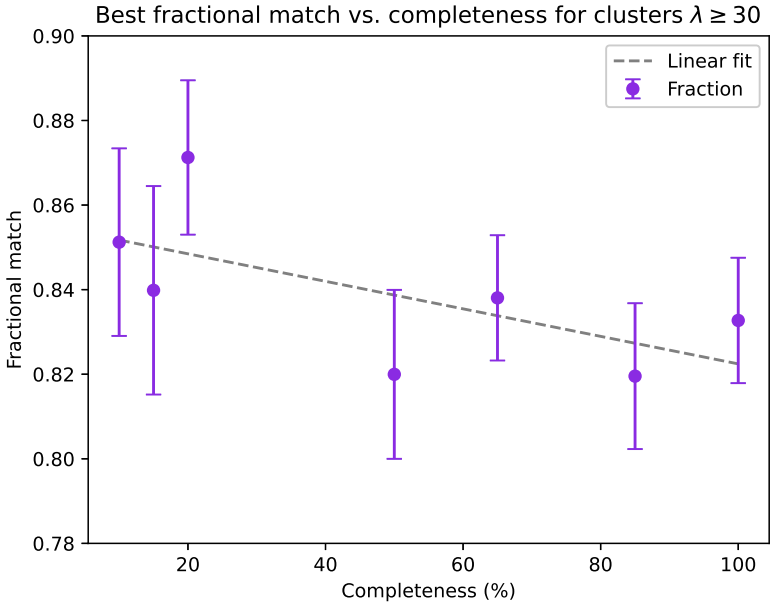


Figure 3.4: The mean best fractional match of clusters with $\lambda > 30$ as a function of completeness. The dataset was sampled 10 times per completeness value. Errors are calculated as SEM. A linear fitting to the results is shown as a grey dashed line.

3.4 Cluster center recovery

In addition to the best fraction magnitudes, the membership-based matching can be investigated by studying how well subcomponent centers have been recovered by the group finding algorithm. Large offsets in the component centers can imply poor recovery of a significant fraction of members, inaccurate cluster configuration, or the presence of meaningful noise that is not spatially symmetric.

The Uchuu simulation halo catalog provides the central galaxy (Brightest Cluster Galaxy; BCG) as the halo center. For the detected clusters, however, choosing the brightest galaxy as the cluster center yields issues particularly in noisy cases where bright field galaxies or lone haloes are present and may poorly represent the true central region. Therefore, two other methods for cluster center estimation are utilized: the plane-of-sky mean and median centers. It is worth noting that numerous more robust cluster center determination methods are available, such as velocity weighted centers or magnitude-based methods. Nevertheless, for the inelaborate purposes of this assessment of center matching, mean and median values of member coordinates are adequate estimates.

Figure 3.5 illustrates an example of comparison between a detected subcomponent and corresponding halo from the simulation catalog, along with the position of the BCG and the mean and median centers. Members in agreement, misclassified members,

and undetected members are presented. Inspection of this result reveals that missed members are predomenantly significantly outside the main component, but a notable amount of noise prevails in the detected cluster, mainly in the form of lone haloes. Both the median and mean are closely matching the same values for the true halo, while it is apparent that the choice of center definition indeed skews the center even within the known halo.

To study how the cluster center agreement develops with reduced completeness, a procedure of sampling is followed similarly to Section 3.1. Analysis is again limited to haloes of known richness $\lambda > 30$. At each completeness level, sampling and running `mclust` on the completeness-reduced subsample is repeated ten times due to the limited number of rich subhaloes in the dataset. Figure 3.6 shows how the separation between the mean and median centers of the matching detected cluster and the given central galaxy of the halo changes with completeness. As expected, agreement improves significantly with increasing completeness, but notably the trend stabilizes already at $\gtrsim 50\%$ for both center methods to values of ~ 620 kpc and ~ 410 kpc for the mean and median centers, respectively. At all completenesses, median centers agree more closely with the central galaxy compared to the mean.

The central matching is affected both by noise and the performance of the group finding algorithm, as well as pure sampling effects. To assess how much sampling influences the central matching, the process is repeated by drawing samples at the same completeness levels as above, but instead directly from the simulation member catalog, so substructures retain their classification information. The same richness cut of $\lambda > 30$ is made, and mean and median centers are calculated from the completeness-reduced groups. These are then again compared to the given central galaxy position. Figure 3.7 presents the results of this process, showing similar behaviour where matching improves and stabilizes to approximately constant above completeness $\gtrsim 50\%$. Here, the median center matches reach values of ~ 130 kpc, while mean centers stabilize at ~ 160 kpc.

The magnitude of the center separation sets a lower limit for the accuracy of cluster center recovery with noise included. While at lower completenesses the separation is almost an order of magnitude larger with noise included, the difference decreases at significantly above 20% completeness.

3.5 Substructure velocity dispersion and virial mass

The velocity dispersion is one of the most commonly used dynamical parameters used to study galaxy clusters. For this reason, the completeness-dependence of the Gapper

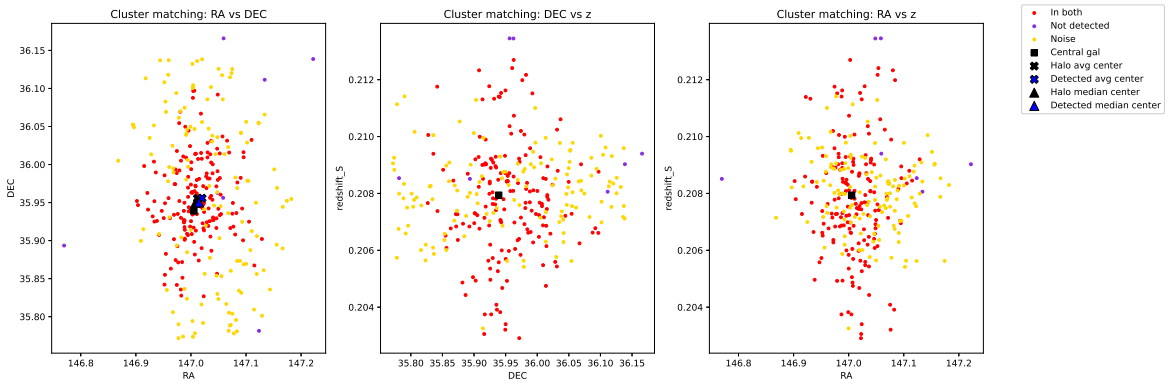


Figure 3.5: Clustering analysis results of one subhalo in the Uchuu dataset and its best-matching group detected by `mclust`. Left: ra [°] vs. dec [°]. Middle: dec vs. z_S. Right: ra vs. z_S. Members present in the halo and the detected group are red, members in the halo but not classified together are violet, and noise (i.e. galaxies classified with the halo but not its real members) is yellow. A black square denotes the given position of the halo central BCG. Coordinate average centers are denoted with an X, filled with black for the known halo and blue for the best-matching detected group. Similarly coloured, triangles represent the median centers.

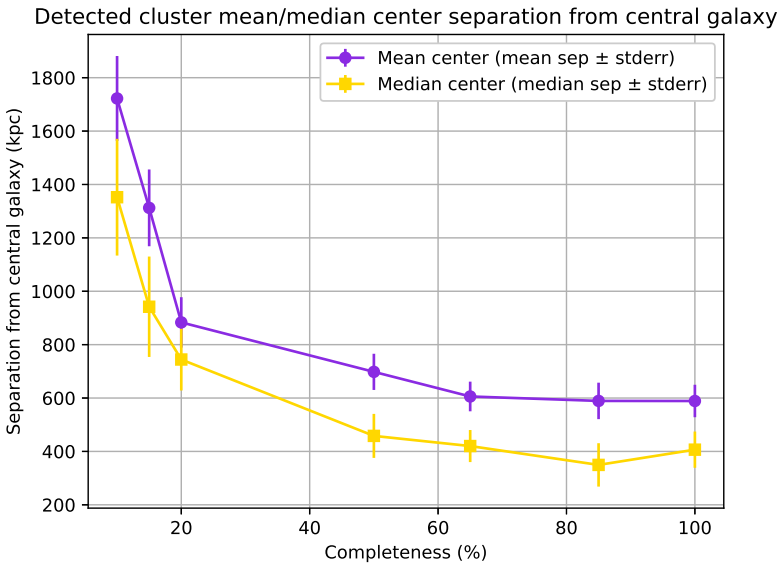


Figure 3.6: Separation in kiloparsecs between detected group mean (violet) and median (yellow) centers and the given central BCG of the halo as a function of completeness. Errors of both center estimators are calculated as SEM. Calculations are limited to subclusters with known $\lambda > 30$, and ten samples have been drawn from the Uchuu dataset at each completeness level.

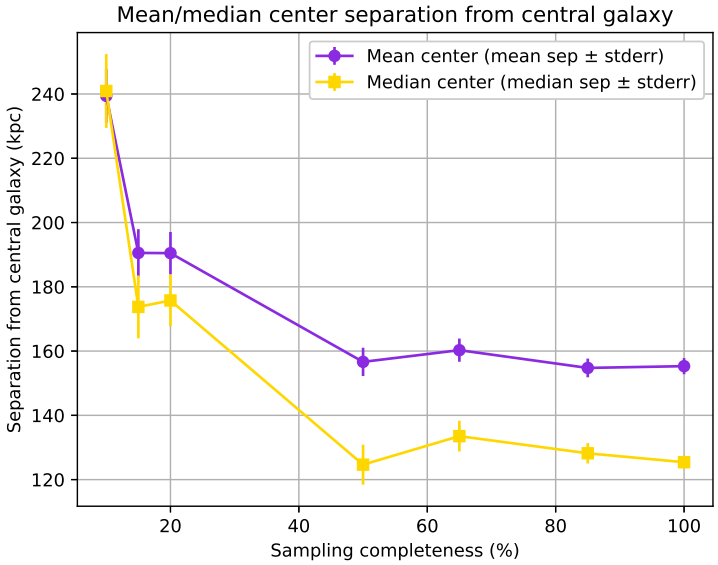


Figure 3.7: Separation in kiloparsecs between completeness-reduced group mean (violet) and median (yellow) centers and the given central BCG of the halo as a function of completeness. Errors of both center estimators are calculated as SEM. Calculations are limited to subclusters with known $\lambda > 30$, and 100 samples have been drawn from the Uchuu dataset at each completeness level. Sampling is done directly from the known clusters in the simulation halo catalog.

velocity dispersion σ_{G}, defined in Section 1.2.2, is investigated.

The behaviour of the velocity dispersion is first studied by examining how sampling affects σ_G measurements. Samples at a given completeness are drawn directly from the simulation halo member catalog, in order to restrict to only sampling effects. For each completeness threshold, 100 samples are generated and σ_G is calculated for subgroups satisfying $\lambda > 30$. The Gapper velocity dispersion was chosen due to its robustness against contamination (Beers et al., 1990), as the present simulation applies maximized noise. The Gapper velocity dispersion is hence expected to represent the velocity structure more accurately. The mean ratio between σ_G of subgroups in the sampled set and the known complete subhaloes are then calculated, and results are shown in Figure 3.8. Results show that values are within 1% of σ_G of the complete subhaloes at all completenesses, but uncertainty is significantly reduced at 50% completeness and above.

Then, the velocity dispersion measurements of the `mclust`-identified substructures, $\sigma_{\mathrm{G,det}}$, are calculated and compared with $\sigma_{\mathrm{G,total}}$, the Gapper velocity dispersion of the known members of the matched halo. As previously, ten samples are generated from the simulation set per completeness level. Membership cleaning is done on the detected groups using the `Clean` routine by Mamon et al. (2013), which rejects of members not fulfilling the conditions $R < r_{\mathrm{vir}}$ and $|v/\sigma_{\mathrm{LOS}}(R)| < 2.7$, where

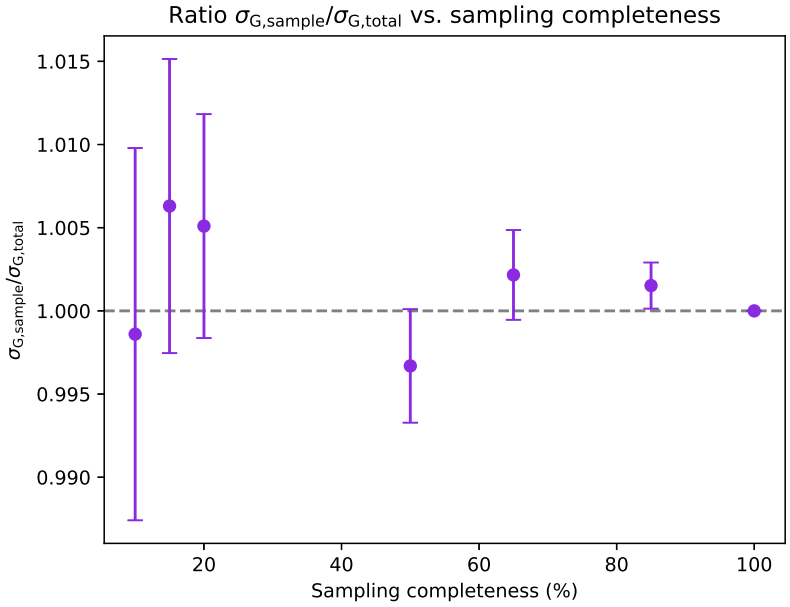


Figure 3.8: The ratio between sample Gapper velocity dispersion, $\sigma_{G,sample}$, and total halo velocity dispersion, $\sigma_{G,total}$, as a function of completeness. Sampling is done from the known clusters in the simulation halo catalog, hence with no noise included. The horizontal dashed line shows $\sigma_{G,sample}/\sigma_{G,total} = 1$. Errors are computed as SEM.

r_{vir} is the virial radius of the group, v is the member's velocity, R is the projected radius, and σ_{LOS} is the line-of-sight velocity dispersion estimated in the routine by assuming an NFW model. The velocity dispersion is then calculated including only the cleaned members. Figure 3.9 shows the cleaned ratio $\sigma_{G,det}/\sigma_{G,total}$ as a function of completeness. Results indicate velocity dispersion is consistently underestimated, with $\sigma_{G,det}/\sigma_{G,total} = 0.79 \pm 0.01$ at 100% completeness. Similarly to findings made for fractional matching and cluster center recovery in Sections 3.3 and 3.4, respectively, results also do not evolve significantly above $\sim 50\%$ completeness. Instead, the ratio remains within the range $0.78 < \sigma_{G,det}/\sigma_{G,total} < 0.80$. Comparing with Figure 3.8 additionally confirms that this gap from the total velocity dispersion is not addressed by sampling effects.

One possible contribution to the seen underestimation may follow from missed component members due to aperture definition effects. Therefore, an additional procedure is added to the component selection and cleaning process. First, for each `mclust`-detected subcomponent in a sample, a minimum volume ellipsoid encompassing all members is defined (details in Appendix A). An example from the 100% completeness sample is shown in Figure 3.12a. Then, the ellipsoid is placed back into the whole sample, and all galaxies falling into the volume are selected and defined as the new detected component. In the example, this would mean that the galaxies marked as blue

in Figure 3.12b would be added to the detected component for further steps. Only after this additional volume selection are the detected subclusters cleaned using `Clean`, and the Gapper velocity dispersions are calculated as described above. The velocity disperson ratios $\sigma_{\mathrm{G,det}}/\sigma_{\mathrm{G,total}}$ of the volume-selected and cleaned subcomponents are included in Figure 3.9, in comparison with results without the additional processing. Behaviour at $c \leq 20\%$ is similar, and values lie within the SEM of each other. At higher completeness, however, the additional volume selection improves the ratio, bringing the value up to $\sigma_{\mathrm{G,det}}/\sigma_{\mathrm{G,total}} = 0.85 \pm 0.01$ at 100% completeness. This could be partially explained by the addition of new members near the volume boundary, which increases the dispersion.

The effects of substructure removal are also tested on the mass of clusters by similarly comparing virial mass estimates for the detected groups to halo virial masses supplied in the Uchuu simulation catalog for groups satisfying $\lambda > 30$. Detected group virial mass, $M_{\mathrm{vir,det}}$, is calculated using Equation 1.5, and the virial radius computed during membership cleaning. Similarly to the σ_G analysis, the virial mass calculations are also repeated for groups with the additional volume selection step. The obtained virial mass estimates of the detected subcomponents are in the range $1.31 \times 10^{13} M_{\odot}/h$ to $3.31 \times 10^{14} M_{\odot}/h$, while true subhalo virial masses reported in the catalog range from $3.27 \times 10^{13} M_{\odot}/h$ to $3.29 \times 10^{14} M_{\odot}/h$. The ratios between the detected group virial mass and halo virial mass in both cases are illustrated in Figure 3.10. At 100% completeness, the value reached by directly applying `Clean` on the matched components is $M_{\mathrm{vir,det}}/M_{\mathrm{vir,halo}} = 0.847 \pm 0.106$. Components first defined using the minimum volume ellipsoid reach $M_{\mathrm{vir,det}}/M_{\mathrm{vir,halo}} = 0.996 \pm 0.117$. The virial mass estimate is again underestimated by the `mclust`-detected groups at $c \geq 65\%$, while staying within $\sim 20\%$ of 1. The volume-selected groups instead agree with $M_{\mathrm{vir,det}}/M_{\mathrm{vir,halo}} = 1$ within 10% in the high-completeness regime, but it is worth noting that all $M_{\mathrm{vir,det}}$ measurements are within the uncertainties of the other method.

In the low-completeness range $10\% \leq c \leq 20\%$, virial mass estimates overshoot in comparison to the simulation haloes for both group selection procedures, with $M_{\mathrm{vir,det}}/M_{\mathrm{vir,halo}} > 1.7$ at 10% completeness. This behaviour is most likely explained by overmerging seen with decreasing completeness, as illustrated in the example in Figure 3.3 and the external metrics ARI and V-measure in Table 3.1.

In regard to addressing the influence of substructure on the $\sigma_v - M$ relation, the obtained subcluster Gapper velocity dispersions and subhalo virial masses are placed in comparison with previous literature findings on the relation from Munari et al. (2013). Figure 3.11 illustrates this comparison at each completeness level, with and without volume selection. Results in both cases suggest that massive subclusters are more drastically affected by the underestimation of the velocity dispersion σ_G, with

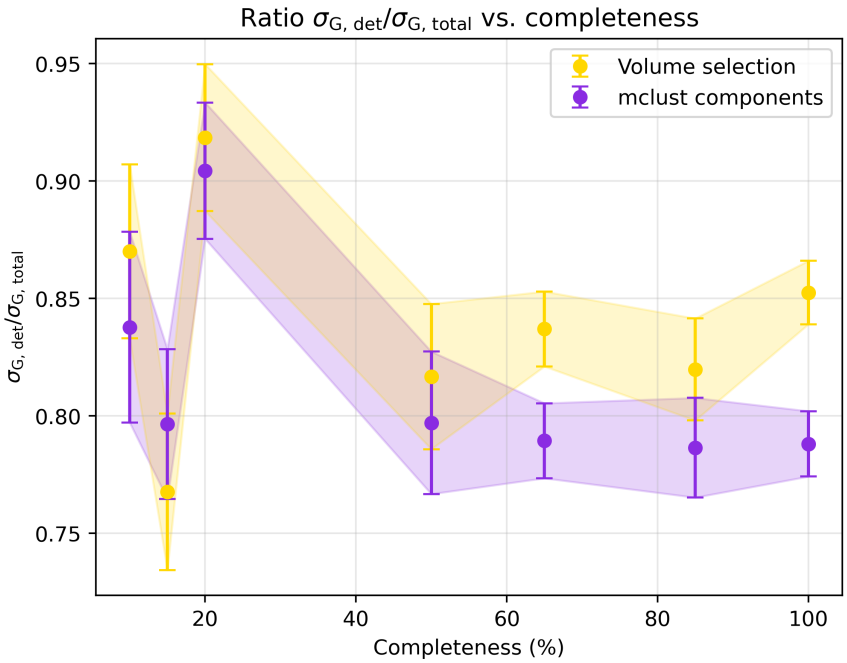


Figure 3.9: Ratio of the Gapper velocity dispersion of detected subclusters and their matched halo's total members, $\sigma_{G,det}/\sigma_{G,total}$, as a function of completeness. The `mclust`-detected components cleaned with `Clean` are shown in violet, and the components with an additional volume selection step are shown in yellow. Errors are computed as SEM, and highlighted within the coloured regions.

values agreeing within uncertainties of the literature relation at lower masses, but diverge with $M_{vir} \gtrsim 10^{14} M_{\odot}/h$. The volume-selected components approach the relation more closely with increasing completeness in comparison to components without the additional processing step.

3.6 Influence of Gaussianity

Relaxed, and hence more Gaussian, clusters are expected to have lower velocity dispersions, while more disturbed clusters with non-Gaussian distributions tend to have increased σ_v. Therefore, the influence of Gaussianity on subcomponent detection and the observed underestimation of σ_v is tested using the Anderson-Darling (A-D) test (Anderson & Darling, 1952). The A-D test has been determined, for example by Hou et al. (2009), to be an effective statistical tool for distinguishing between relaxed and disturbed clusters. It is used here to compare the A-D scores of the detected subcomponents to the score of the known true subhalo members from the truth-valued catalog.

To make this evaluation, A-D test scores are compared to critical value estimates at the 5% significance levels. If the A-D test statistic, A^{*2}, exceeds the critical value,

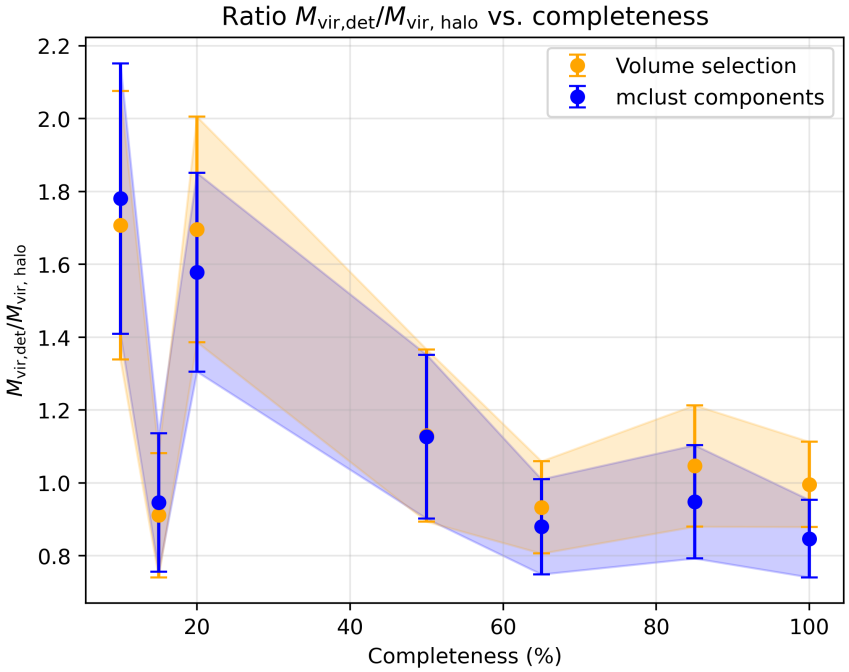
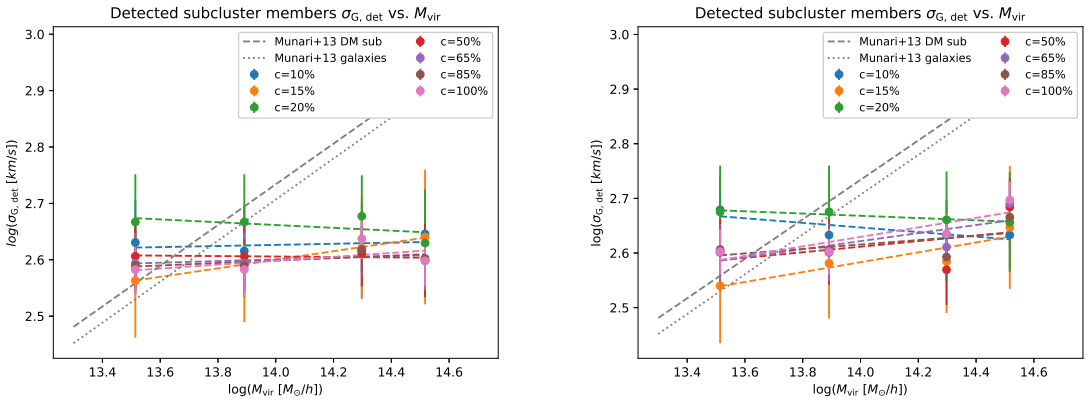
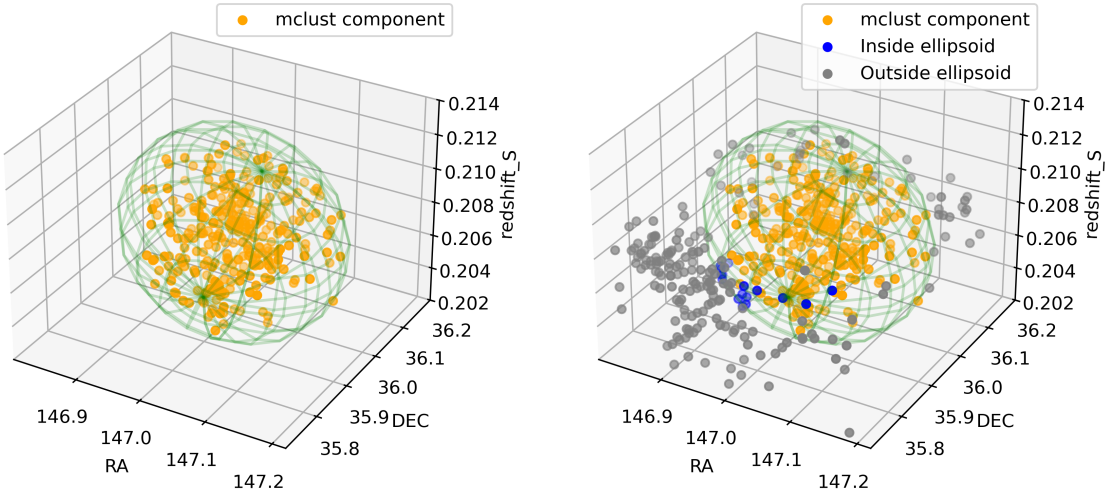


Figure 3.10: Ratio of virial mass estimate of detected subclusters and the matched halo, $M_{\mathrm{vir,det}}/M_{\mathrm{vir,halo}}$, versus membership completeness. The `mclust`-detected components cleaned with `Clean` are shown in blue, and the components with an additional volume selection step are shown in orange. Errors are computed as SEM, and highlighted within the coloured regions.



(a) `mclust` components + `Clean`. **(b)** `mclust` components + volume selection + `Clean`.

Figure 3.11: Velocity dispersion (kms^{-1}) of detected substructures as a function of the halo virial mass M_{vir} ($10^{14}M_\odot/h$) for components with $\lambda > 30$, separated by color for each completeness threshold. Virial mass values are applied from the Uchuu halo catalog. Dashed and dotted grey lines illustrate the scaling relations obtained by Munari et al. (2013) for the dark matter only simulation subhaloes and AGN simulation galaxies, respectively. Subfigure (a) shows the $\sigma_v - M_{\mathrm{vir}}$ scaling relation on the cleaned `mclust`-detected components, while additionally volume-selected components are shown in (b).



(a) Ellipsoid fit.

(b) Selection from all galaxies.

Figure 3.12: Illustrative example of the minimum volume ellipsoid fitted around an `mclust`-detected subcomponent. Positions plotted as plane-of-sky positions, ra [°] and dec [°], and observed redshift. (a) shows the members classified as a component (orange) by `mclust` and the corresponding volume. (b) shows the ellipsoid placed into the whole sample, including the `mclust` component (orange), galaxies outside of the ellipsoid (grey) and galaxies located within the ellipsoid that were not assigned to the group by `mclust` (blue). Plots created using a helper function from `Ellipsoid Fit` (Sandu, 2021, see Appendix A).

the null-hypothesis can be rejected at the given significance level. In other words, the component is then considered non-Gaussian.

Table 3.2 shows the A-D test result as fractions of non-Gaussian substructures across detected components with $\lambda > 30$, again from the 10 samples constructed per completeness level. Results suggest that the true subclusters are less likely to be Gaussian compared to detected groups regardless of the added volume selection step. However, the volume-selected subclusters have slightly higher fractions of non-Gaussian clusters at low completeness ($c \lesssim 50\%$), but the opposite is seen with the full-completeness samples, where `mclust + Clean` processed subclusters have a non-Gaussianity fraction of 0.35, while volume-selected samples have 0.15. True sub-halo members have a non-Gaussianity of 0.50 at 100% completeness. This high non-Gaussianity could be a contributing factor to the issue of high velocity dispersions compared to detections.

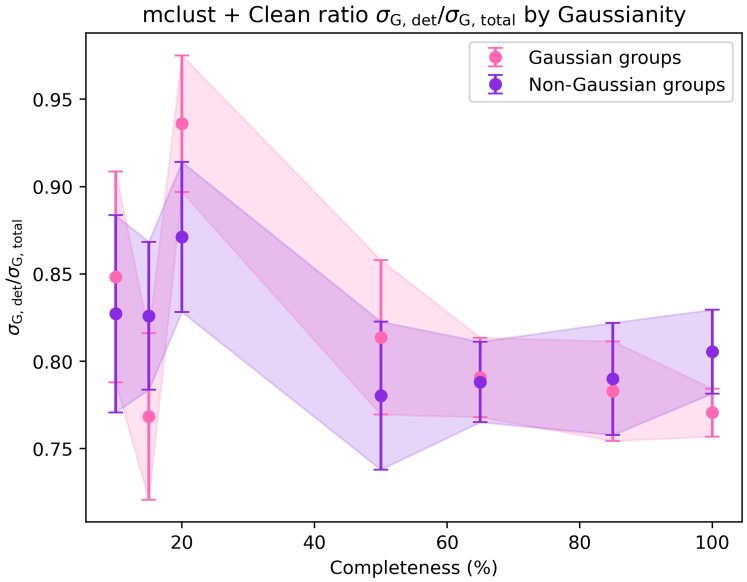
In order to test whether the increased Gaussianity of detections is resulting in the underestimation of velocity dispersion measurements, the $\sigma_{\mathrm{G,det}}/\sigma_{\mathrm{G,total}}$ ratios are again computed while separating the subclusters by Gaussianity of the true groups, as obtained from the A-D test. Figure 3.13 presents these results for detected cluster member selection with and without definition of the ellipsoid. Notably, the Gaussian

Table 3.2: Results of the A-D test applied to the detected clusters with $\lambda > 30$ at different completenesses, calculated across 10 samples per completeness level. The columns show the fraction of subclusters that fail the A-D test at the 5% significance level, and are hence considered to be non-Gaussian (non-G). The final column shows these fractions for the truth-labelled subclusters, also drawn from 10 samples per completeness level.

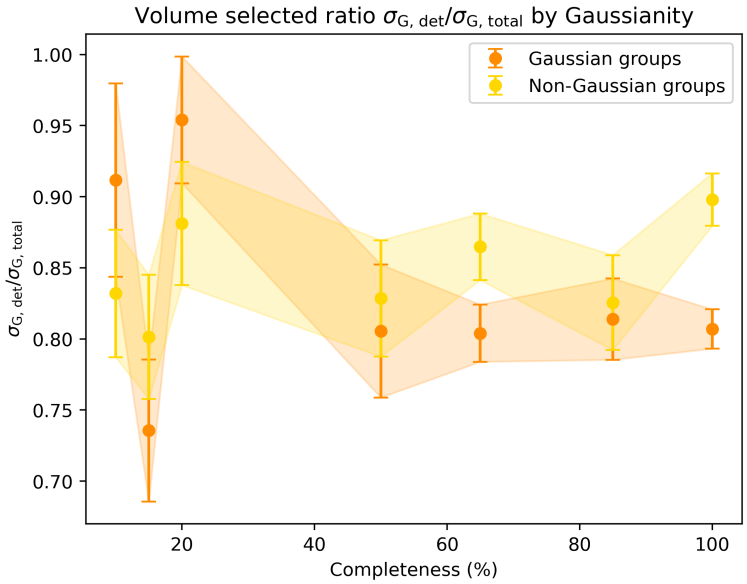
Completeness (%)	Fraction of non-G groups `mclust+Clean`	Fraction of non-G groups Volume selection	Fraction of non-G groups true
10	0	0.0434	0.5938
15	0	0.0513	0.4750
20	0.0513	0.1282	0.4740
50	0.1250	0.1538	0.5000
65	0.1842	0.0571	0.5263
85	0.1250	0.1750	0.5000
100	0.3500	0.1500	0.5000

groups in the aperture-selected groups show similar evolution with increasing completeness compared to subclusters without this step. In the `mclust + Clean` groups, no significant influence of non-Gaussianity is seen on the velocity dispersion ratio. The only divergent trend is seen in non-Gaussian clusters in the volume-selected method, which show increased ratio at higher completeness, and are likely the main contribution to the differences seen in Figure 3.9.

Nevertheless, the results obtained from Gaussianity-based classification of clusters do not show differences significant enough to explain the underestimation of velocity dispersion, and confirm the need of a deeper study.



(a) `mclust` components + `Clean`.



(b) `mclust` components + volume selection + `Clean`.

Figure 3.13: Ratio of the Gapper velocity dispersion of detected subclusters and the true halo's total members, $\sigma_{G,det}/\sigma_{G,total}$, as a function of completeness. (a) shows the `mclust`-detected components cleaned with `Clean` and separated by Gaussianity, with non-Gaussian clusters in violet and Gaussian clusters in pink. Components with an additional volume selection step are shown in (b), where non-Gaussian groups are depicted as yellow and Gaussian groups as darker orange. Errors are computed as SEM, and highlighted within the coloured regions.

4. Discussion and conclusions

This thesis has investigated the characterization of galaxy cluster substructure dynamics from spectroscopic data using group finding algorithms. The question lies at the interconnection between spectroscopic data properties, informatics and algorithm performance, and the scarcely studied subcluster regime.

The spectroscopic redshift completeness in cluster catalogs is addressed first. Optical data of clusters from the DECaLS redMaPPer catalog (Rykoff et al., 2014; Kluge et al., 2024) is enriched with spectroscopic redshift measurements from Data Release 1 of the recent DESI main survey (DESI Collaboration et al., 2026) in order to evaluate improvements on redshift completeness obtained with this strategy. Richness and redshift cuts, as well as data quality cuts to the scale factor and mask fractions, are applied to the redMaPPer clusters in order to prepare a suitable subset for substructure analysis. Members are then matched to DESI DR1 data and enriched with spectroscopic redshifts. This method allows utilizing the new data without challenges imposed by the DESI sampling function that could bias the group finder results.

DESI enrichment improved the spectroscopic redshift completeness distribution of the subset, with most significant contributions in the medium and high completeness ranges. The number of clusters with completeness $N_{z_{\mathrm{spec}}}/N_{\mathrm{gal}} > 30\%$ was improved by 58.8%, while the high completeness range $N_{z_{\mathrm{spec}}}/N_{\mathrm{gal}} > 75\%$ saw improvements of 76.7%. Enrichment also particularly improved the redshift counts per cluster near $N_{z_{\mathrm{spec}}} \sim 20$.

In order to estimate data quality, the residuals between spectroscopic redshift measurements of galaxies for which both redMaPPer and DESI data are available were investigated. Residuals were largely found to be below the critical threshold for influencing cluster velocity dispersions, but a significant dependence on magnitude was detected. At magnitudes of mag ≥ 17, the fraction of logarithmic residuals $r_{\log} > -3$ approaches 10%.

These investigations on spectroscopic data use suggest that the enrichment of redMaPPer clusters in combination with a significantly larger and fainter spectroscopic catalog like DESI is a credible strategy for utilizing novel data. Similar methodology has been used, for example by Myles et al. (2025), for characterizing redMaPPer clusters

and projection effects.

In addition to data quality assessment, another key question discussed in this work is the selection of a group finding algorithm. The compatibilities of four group finding algorithms, the shifting gapper method (Fadda et al., 1996; Lopes et al., 2009), the halo-based method (Tinker, 2021, 2022), the friends-of-friends algorithm (Tempel et al., 2016), and `mclust` (Scrucca et al., 2023), are tested for purposes of substructure detection within rich galaxy clusters. The properties of `mclust` were deemed most suitable after prior assessments of, inter alia, the filtering conditions, underlying assumptions of halo structure, and preliminary performance tests of the four algorithms. Hence analysis was focused on `mclust` and its ability to recover statistics of merging systems.

The algorithm's performance was tested on Uchuu dark matter halo simulation data (Ishiyama et al., 2021), populated using the technique described in Comparat et al. (2020). Evaluating five Cluster Validity Indices of partitions at seven different completeness levels in the range $10\% - 100\%$ revealed that solutions outperform photometric results near $\sim 20\%$ completeness, when compared with a similar study by Sepúlveda et al. (in prep.). However, it is also important to assess how useful and descriptive the metrics are for the datasets in question.

For example, comparing the Davies-Bouldin index results of the photometric data to the results obtained in this work suggests that no significant improvements need to be made on the photometry. There are, however, multiple possible factors influencing these results. Namely, the effects of the richness cuts of $\lambda > 30$ applied to the Uchuu subset here, and the cuts applied by Sepúlveda et al. of $\lambda > 4$, could be investigated further. Alternatively, results could be compared to cluster data with no richness cuts to evaluate the dependence of group finder performance on predefined richness cutoff thresholds, and determine possible dynamical range loss. In this study, focus was placed on the richest subcomponents for the purposes of analysis of dynamical properties.

While the largely similar Calinski-Harabasz index showed better performance for `mclust` compared to both DBSCAN and the other GMM model used on photometry, the reliability of the index is unclear. As discussed in Section 2.4, the Silhouette and DB have been found to be more widely applicable with varying group shapes (Chicco et al., 2025). Even so, both DB and CH are limited by how they define a "good" cluster; they are based on the assumption that good groups are densely populated and separated far from one another. This is rarely the case for intra-cluster substructure, especially with old mergers or clusters that have undergone multiple merging events. Therefore, reliably applying these CVIs requires better understanding of the reference values for the dataset in question. As has been done in this work, computing the inter-

nal CVIs on the truth-labelled data supplies these reference values, but they will not be directly translatable onto observational data with drastically different densities, cluster shapes and richnesses. Further investigation into finding expected values based on data properties, for example, using larger datasets and a wider range of substructure, poses compelling future prospects for characterizing clustering solutions on observations.

External metrics, on the other hand, are better suitable for direct comparison, because they describe how well algorithms replicate their respective intended component solution. They are hence more objective proxies of algorithm performance. Comparison with results from Sepúlveda et al. confirms a need for improvements from photometry. Both the Gaussian mixture modeling algorithm and DBSCAN are outperformed with spectroscopic substructure identification at $15\% - 20\%$ spectroscopic completeness when comparing to external CVIs.

It is not expected that spectroscopic analysis would outperform group finding from photometry regarding all parameters. This work has aimed to characterize the improvements that the addition of spectroscopic redshift data can bring to substructure analysis, and found that improvements are made particularly with the purity and homogeneity of obtained partitions.

Contrastingly, limitations of `mclust` were encountered when recovering dynamical properties of subcomponents in the partition. Both the Gapper velocity dispersion, σ_G, and virial mass, $M_{\rm vir}$, of the detected groups were underestimated at completeness levels $c \geq 50\%$, and comparison to literature $\sigma_v - M_{\rm vir}$ scaling relations suggests that massive components are affected more drastically. Membership cleaning using the `Clean` routine by Mamon et al. (2013) is done to rule out trivial resolving of the issue through noise, but the trend persists on cleaned groups. This presents a considerable challenge to the applicability of `mclust` as is, and demostrates the main disadvantage of the method. Sepúlveda et al. also find that their Gaussian mixture modeling method performs best at detecting central structures of the components, which agrees with verdict made here. Assumption of Gaussianity could introduce biases in detections as membership may be artificially approaching Gaussian distributions, leading to underestimation of the velocity dispersion.

For this reason, the influence of Gaussianity of subcomponents is investigated further using the A-D test, and structures are separated into Gaussian and non-Gaussian. It is confirmed that substructure detection shows a significantly higher number of Gaussian clusters compared to the true halo members. Re-calculation of velocity dispersions with this separation, however, does not explain the underestimation of σ_v, and rules out the straightforward explanation for the problem. One future prospect for an explanation for these findings would be a further study into the velocity profiles of subcomponent members with radial velocity from the center, which would reveal in

detail the presence of noise within the redshift range of the subcomponents.

Overall, results and examinations of the presumable roots of the issue still do not alleviate the incomplete recovery of dynamical properties with `mclust`. This formulates the framework of a new research problem, namely that the $\sigma_v - M_{\rm vir}$ relation for substructure cannot be measured in the same manner as previously assumed. Hence, effective utilization of velocity dispersion measurements for characterizing infalling structures yet requires re-calibration of the $\sigma_v - M_{\rm vir}$ relation.

On the other hand, the process of substructure detection has demonstrated the specific requirements in the field for efficient utilization of informatics tools and high-performing algorithms to develop further understanding of machine learning methods for large-scale substructure finders. This gap in the field has already been a point of discussion within the last decade (see Yu et al., 2015, de Los Rios et al., 2016, Tempel et al., 2017), and remains a current topic of research in cluster studies today. All of these outlined questions will continue to grow in importance with the emergence of spectroscopic surveys larger than ever before with DESI (DESI Collaboration et al., 2026) and the upcoming 4MOST operations (de Jong et al., 2019), among others.

Appendix A. Minimum volume ellipsoid

This section describes the methodology for defining the minimum volume ellipsoids introduced in Section 3.5. The process begins with selecting the members of a given subcomponent resulting from an `mclust` run. Then, the parameters of the minimum volume encompassing ellipsoid, also known as the Löwner-John ellipsoid, are determined using the `outer_ellipsoid_fit` function in the `Ellipsoid Fit` repository by Sandu (2021).

The function is given the 3-dimensional member data, to which an ellipsoid is then iteratively fitted by decreasing the volume in steps while consulting the Mahalanobis-like distances of points. If points lie outside the volume, the shape is corrected, and the iteration begins again, until convergence. The procedure then outputs a shape matrix $\mathbf{A}$ and centroid $\mathbf{c}$, which can be used to describe the surface of the ellipsoid as

$$(\mathbf{s} - \mathbf{c})^T \times \mathbf{A} \times (\mathbf{s} - \mathbf{c}) = 1, \tag{A.1}$$

where $\mathbf{s}$ is a point on the ellipsoid surface. Now, these parameters are used on the whole sample in order to find all galaxies within the ellipsoid. That is, to find members missed by `mclust` and to include them in further analysis. The check follows directly from Equation A.1:

$$(\mathbf{x} - \mathbf{c})^T \times \mathbf{A} \times (\mathbf{x} - \mathbf{c}) \leq 1, \tag{A.2}$$

where $\mathbf{x}$ is the 3-dimensional position vector of a given galaxy. Any point satisfying Equation A.2 is located within the ellipsoid, and is hence added to the detected cluster.

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